

Design Prototype

Jeremiah Dados, Dmytro Stavskyi, and Luis Wong

1. Effort breakdown

Week	Jeremiah Dados		Luis Wong		Dmytro Stavskyi	
	Tasks	Time spent (hours)	Tasks	Time spent (hours)	Tasks	Time spent (hours)
5/10/2025 - 11/10/2025	Project roadmap design	2	Reading papers that explain the purpose and use cases of the DTCP	2	Researching and understanding the components (DDS, Balun, etc.) and overall structure and design of the original prototype	3
	High-level system architecture research	5	Reading papers and making slideshow for the ADS1299 for weekly meeting w/ professor	5	Researching and synthesizing papers and other sources about the possible platform (ADS1299) for the EMG recorder for the weekly stakeholder meeting	4
	Introducing Luis and Dmytro to the lab space	3	Getting introduced to the lab space and soldering old PCB	3	Lab tour and practice with the original prototype	3
12/10/2025 - 18/10/2025	Researching HF PA solutions (literature review)	3	Researching HF PA solutions (literature review)	5	Researching EMG signals and how they are usually processed and captured	3
	HF PA simulations: class- A	3	HF PA simulations: Class-A	2	Researching how ADS1299 and the MCU (MSPM0C1104) can interact with each other (SPI and pin connections)	3
	Researching MCU options and creating MCU wiring diagram	4	Creating and presenting slideshow for PAs for weekly meeting w/ professor	3	Setting up Altium, examining the Altium schematics of the original prototype and setting up the library for the ADS1299 chip	4
19/10/2025 - 25/10/2025	Researching HF PA solutions (literature review)	3	Researching HF PA solutions (literature review)	7	Synthesizing information for the and creating the weekly stakeholder presentation about the ADS characteristics	4
	PA circuit topology design	3	PA circuit topology design	3	Researching the use of ADS1299 in EMG and other signal processing systems	4
	MCU circuit topology design	1	HF PA simulations: Class-E	3	Preliminary EMG Recording circuit design	2
	Preliminary PCB design	3	Improve readability and design of GitHub page.	1		
26/10/2025 - 1/11/2025	HF PA design iterations	3	Researching HF PA designs.	5	Working on ADS1299 Altium Schematic	4

	Finishing the system design.	2	Researching for and designing Altium schematic for Class E PA	3	More researching how ADS1299 is utilized in the research papers and specific schematics	4
	Preliminary Altium schematic design.	5	Running LTspice simulations to test functionality with real parts.	3	Reviewing Altium schematics of other components	2
2/11/2025 - 8/11/2025	Altium PCB layout iterations	5	Finding alternative parts with high Q for inductors and capacitors. Look for alternative FET with smaller package size.	5	Finishing ADS1299 Altium Schematic	4
	Altium schematic design iterations.	3	Reading papers to better understand Class E Amplifiers.	3	Reviewing Altium Schematics	4
	First prototype PCB design reviews.	2	Review Altium schematic and rerun simulations according to feedback by Han.	3	Research ADC on ADS1299 and how delta-sigma ADCs work overall	2
9/11/2025 - 15/11/2025	First prototype PCB design reviews.	2	Installing and setting up CCS in order to begin writing firmware code.	2	Setting up and testing the MCU	2
Total Time (hours)		52		58		52

Table 1: Breakdown of tasks for each team member. Much of this work is also logged in our [GitHub commit history](#).

2. Evidence of soundness

a) Theoretical Background and Prior Art

The main background research involved in this project deals with RF circuit design and EMG signal recording/processing.

EMG Theory

The signal that we are analyzing in this system is of the electromyography (EMG) type, which captures the electrical activity of muscles. The magnitude of the signal, before amplification, is in the mV range and has a bandwidth of up to 500 Hz [6]. Some possible interferences to the signal include crosstalk (e.g., if EMG is measured near the heart, it can be interfered with by the ECG signal) and coupling with the power supplies [6].

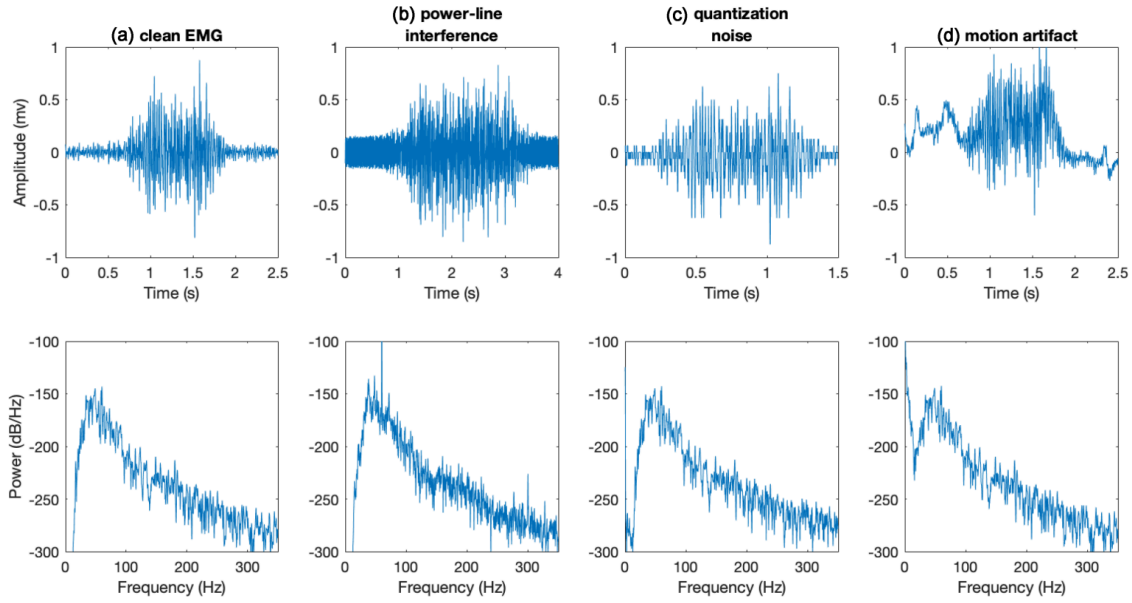


Fig. 1. Sample EMG signals in the time and frequency domains for (a) clean EMG and EMG contaminated with (b) power-line interference, (c) quantization noise, and (d) motion artifact.

Figure 1: EMG signals plotted in the time and frequency domains taken from [6].

The signal is transmitted to the system using needle electrodes. Considering the small amplitude and the susceptibility to noise of EMG signals [6], the important characteristics for the device processing this signal include its Common-mode rejection ratio (CMRR), which quantifies the ability of device to reject signal common to both the positive and negative input terminals and Signal-to-noise ratio (SNR), which is the ratio of signal power to noise power.

One setup using ADS1299, which is the same chip we decided to use as the analog front-end (AFE) in the recording system, is described in the "Design of low-noise multi-channel active surface electromyography acquisition equipment" paper [7]. Before being processed by the AFE, the signal is filtered and amplified. Afterwards, the signal is processed by ADS1299 and then transmitted to the control chip.

Overall, an EMG recording can be divided into four stages: sensing, amplifying, filtering, and ADC. The sensing part is left to the electrodes, which act as an external input to the system. ADS1299 has an internal amplifier with programmable gain of up to 24x, and while most setups described in the journal articles have amplification in the hundreds, considering that the ADC system inside the chip has a resolution of 24 bits, we decided that this gain provided by the internal amplifier will be sufficient. This will also not introduce an additional component(s), so the front-end acts as an actual front-end. The sampling (data) rate of the ADS ranges from 250 SPS (samples per second) to 16 kSPS, so, assuming that 500 Hz is indeed the top limit of the signal, the sampling rate should be at least 1000 SPS to avoid aliasing, but regardless, the sampling rate is programmable.

RF PA Classes and Theory

One of the main goals of the device is to present a wireless neurostimulation platform powered via Differential Tissue Coupled Powering (DTCP) that combines the functionality of various devices for researchers, similar to how the Analog Discovery, made by Digilent, works to consolidate the many devices used by electrical and

computer engineers. The DTCP transmitter is required to deliver differential high-frequency currents (20–25 MHz) into biological tissue while operating from a thin 3.7 V Li-ion battery [4]. With this in mind, our main motivation when choosing a PA topology is to maintain HF frequencies while minimizing power loss and component count in order to keep the device small.

Traditional linear PA classes, such as Class-A, Class-B, Class-AB, and Class-C, operate the active device (the FET) as a current source, never allowing the device to fully saturate. While this typically results in good linearity (an amplifier's ability to produce an output that is faithful to its input) in the case of Classes A, B, and AB, such linearity is unnecessary for our purposes and results in a loss of efficiency as a result of switching losses, especially in high-frequency applications. Therefore, linear PA classes are not suitable for the compact DTCP transmitter.

TABLE II
COMPARISON OF DIFFERENT AMPLIFIERS PRODUCING SINEWAVE
OUTPUT [3]

Class	P_{max}	%	Comments
A	0.125	50	360° conduction angle
B	0.125	78.5	180° conduction angle
C	0.0981	89.6	120.6° conduction angle
	0.0000	100	0° conduction angle
D	0.318	100	uses two devices with 1 A peak current
E	0.0981	100	optimum-50% duty cycle
F	0.159	100	infinite number of resonators
HiQ	0.203	81.8	Efficiency is fundamental frequency output/dc input [12]
S	0.125	100	Ac coupled output

Note: All cases assume ideal transistors: zero saturation voltage, zero saturation resistance, infinite off resistance, and zero switching time.

Class-E amplification offers a more appropriate solution. First introduced by Sokal and Sokal, Class-E circuits operate a single transistor, and it does so as a switch rather than a current source. A properly designed load network ensures that the transistor turns on only when the drain voltage is near zero (Zero-Voltage Switching, ZVS) and the drain current is minimized at turn-off (Zero-Current Switching, ZCS). Under ideal conditions, this eliminates simultaneous high voltage and high current in the device, reducing switching losses dramatically compared to other PA classes [2]. This switch amplifier offers these advantages while also being the least complex of its group. Class C and F amplifiers are subject to complex resonant and harmonic networks, and class D makes use of multiple transistors, introducing additional complexity due to the need for phase matching.

GaNFET versus MOSFET Selection

Device selection significantly affects Class-E performance, particularly at HF frequencies. Silicon MOSFETs, while inexpensive and easy to source, have relatively large output capacitances, slower transition times, and larger package sizes. On the other hand, Gallium Nitride (GaN) FETs offer significant advantages in this frequency range. GaN devices exhibit substantially lower C_{oss} , reduced charge storage, and faster rise/fall times, all while remaining much smaller than their silicon counterparts [5].

b) Empirical evidence

The only part of the design that was developed experimentally so far is the class-E PA circuit, as it was possible to investigate it using SPICE software. The following figures summarize the design process of the Class-E PA:

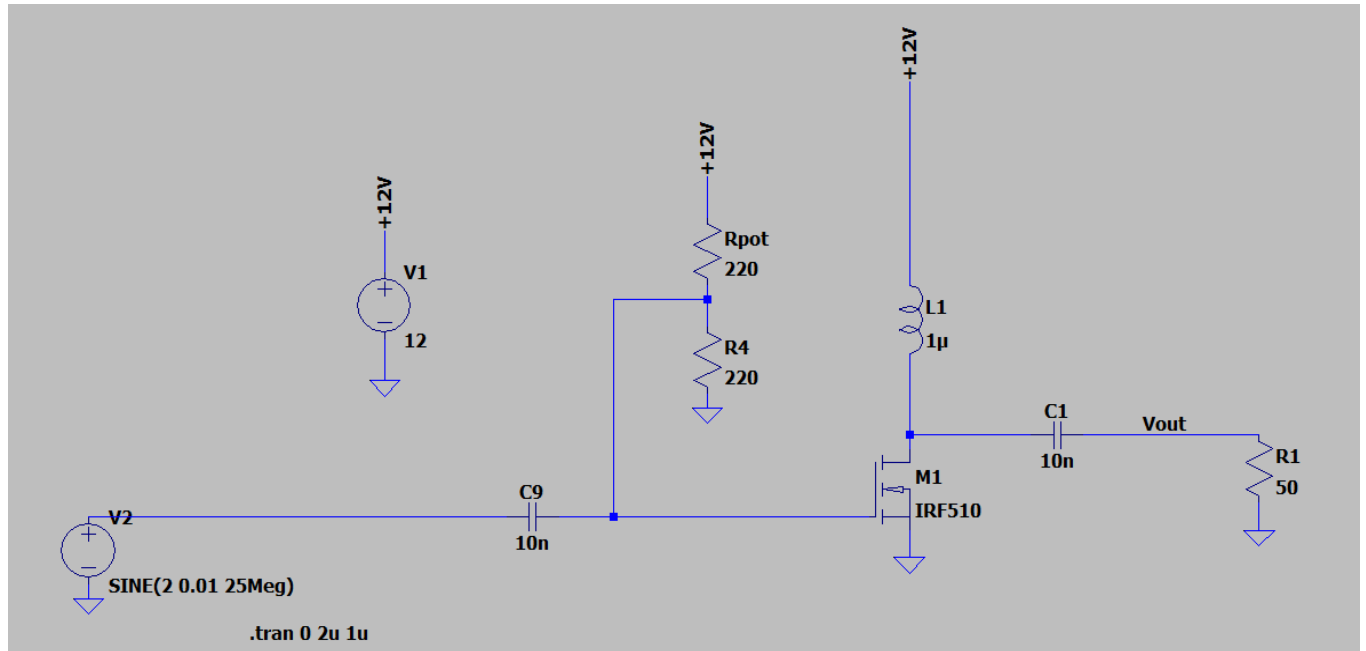


Figure 2: When we began the design of the PA, we started with the simplest one, a Class-A PA (this was before research).

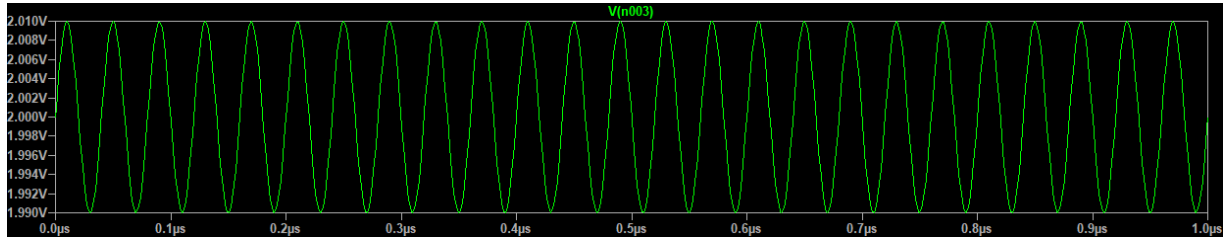


Figure 3: Input signal simulating what would be output from the DDS.

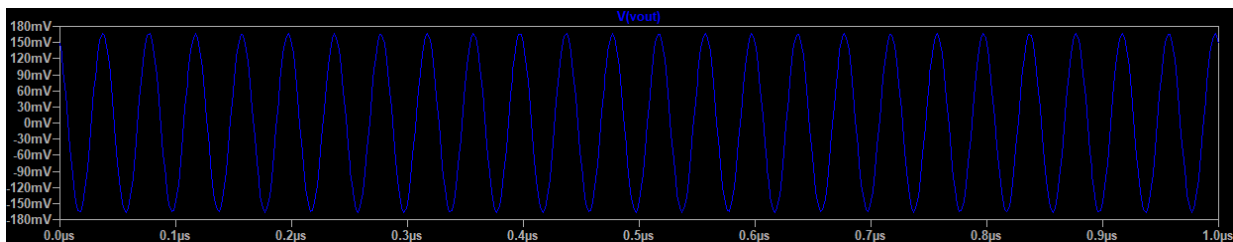


Figure 4: Output signal showing that the signal was amplified.

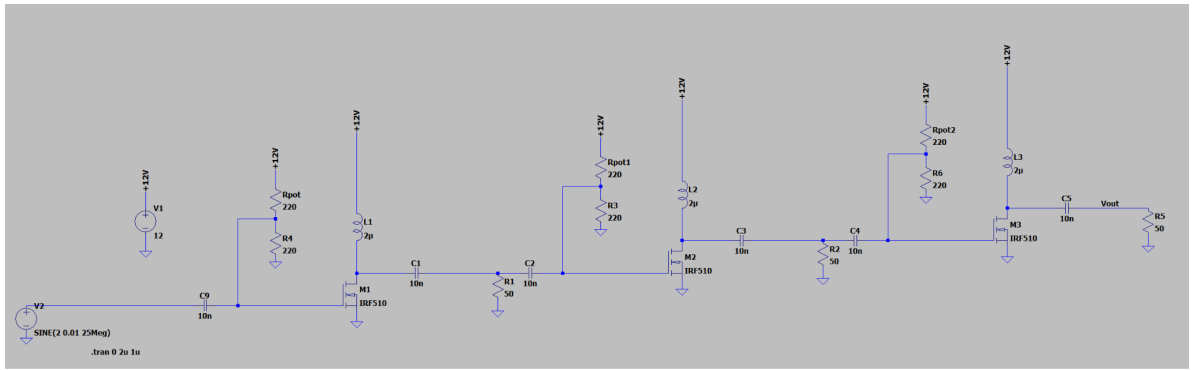


Figure 5: Gain was insufficient, so we decided to put three Class-A amplifiers in series.

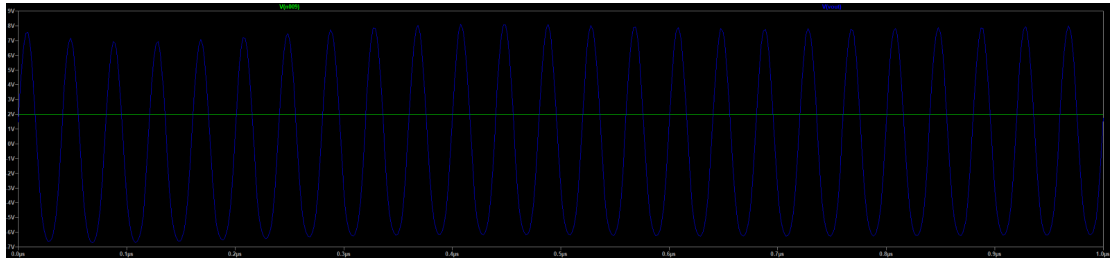


Figure 6: The Signal shows amplification resulting in a signal of ~ 14 Vpp, which is about 27 dBm. The amplitude of the signal oscillated in amplitude but with some tuning; it was dampened somewhat, but not enough to be ideal. If we want to reach the desired power, it would require too many transistors.

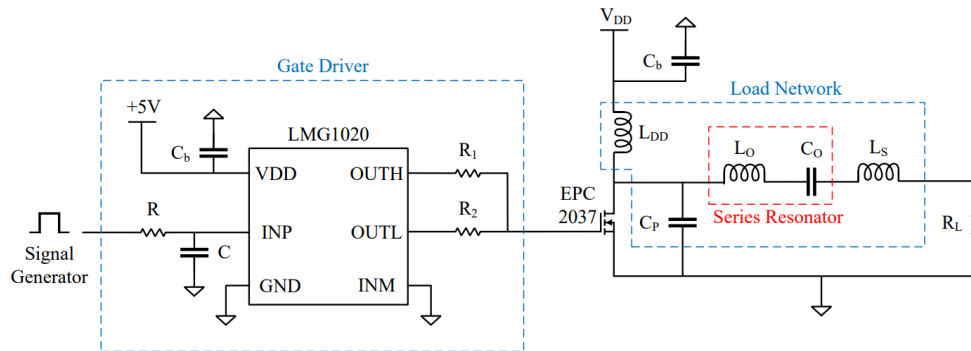


Figure 7: Image and inspiration come from [5] for parts/layout and utilized a calculator (<https://people.physics.anu.edu.au/~dxt103/calculators/class-e.php>) based on formulas found in [1].

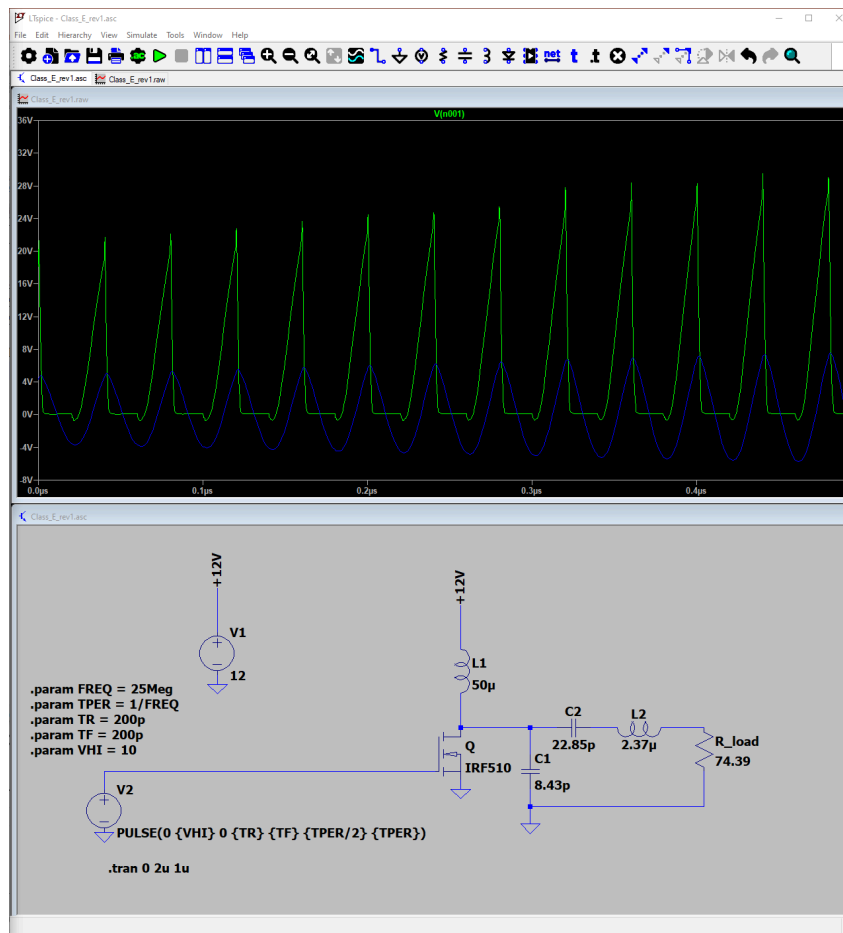


Figure 8: Initial attempt at designing and simulating the Class E PA. Expected a V_{pp} of 24 but got a V_{pp} of 8 instead.

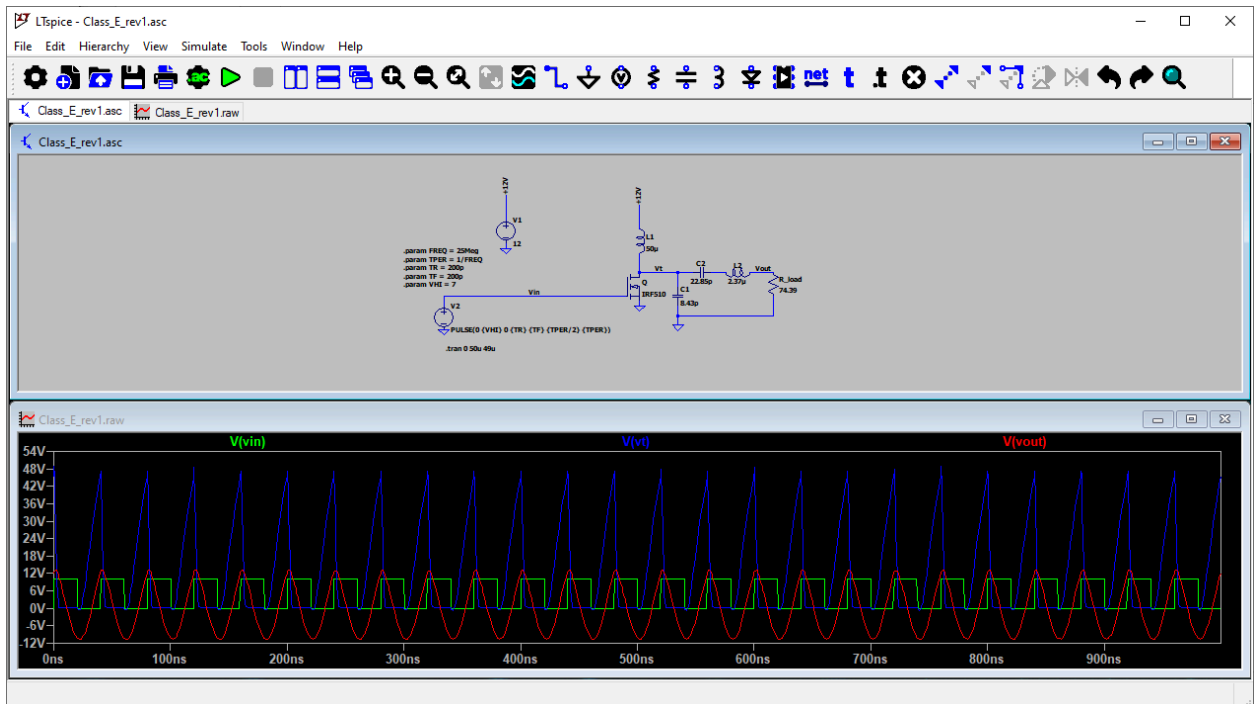


Figure 9: After making the simulation wait 40 microseconds, the result matched the expectation. The circuit needs a little time to warm up.

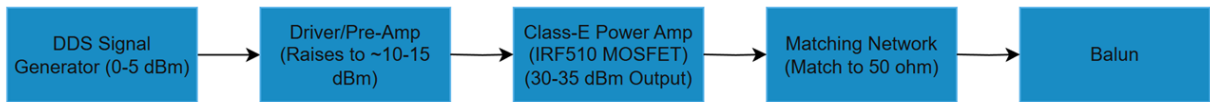


Figure 10: High-level view of PAs' design.

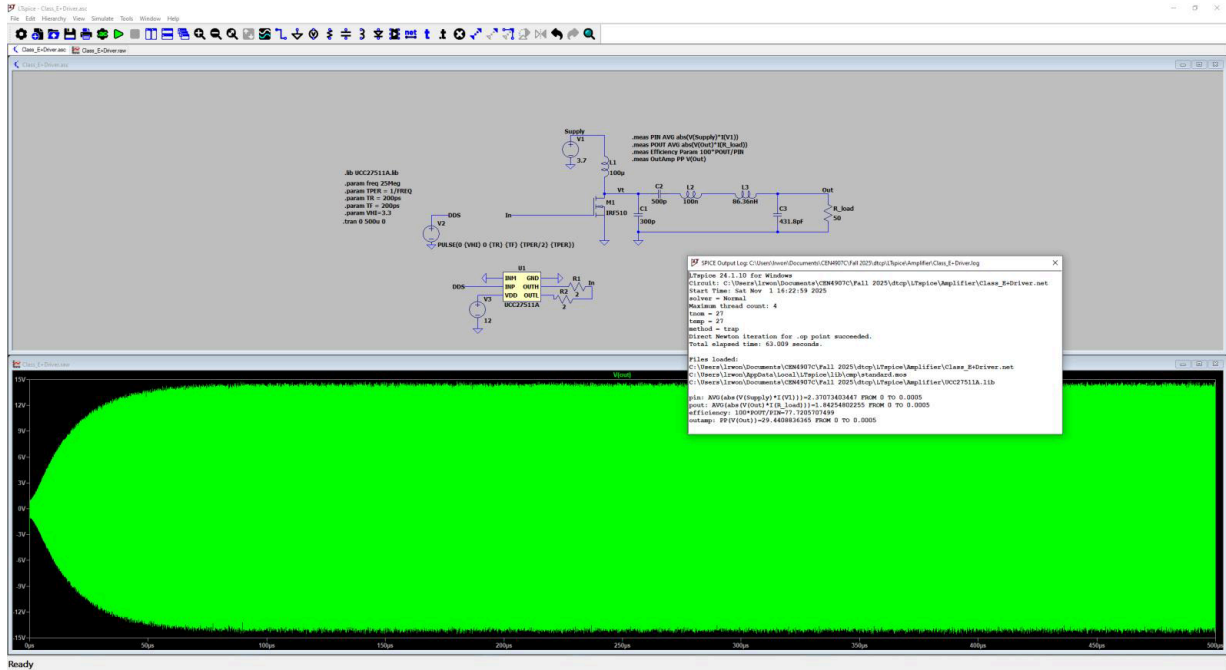


Figure 11: Once the design of the Class-E PA was finished, we needed to simulate it working with the driver and impedance matching network. Here is a simulation showing it working with an approximate 78% efficiency.

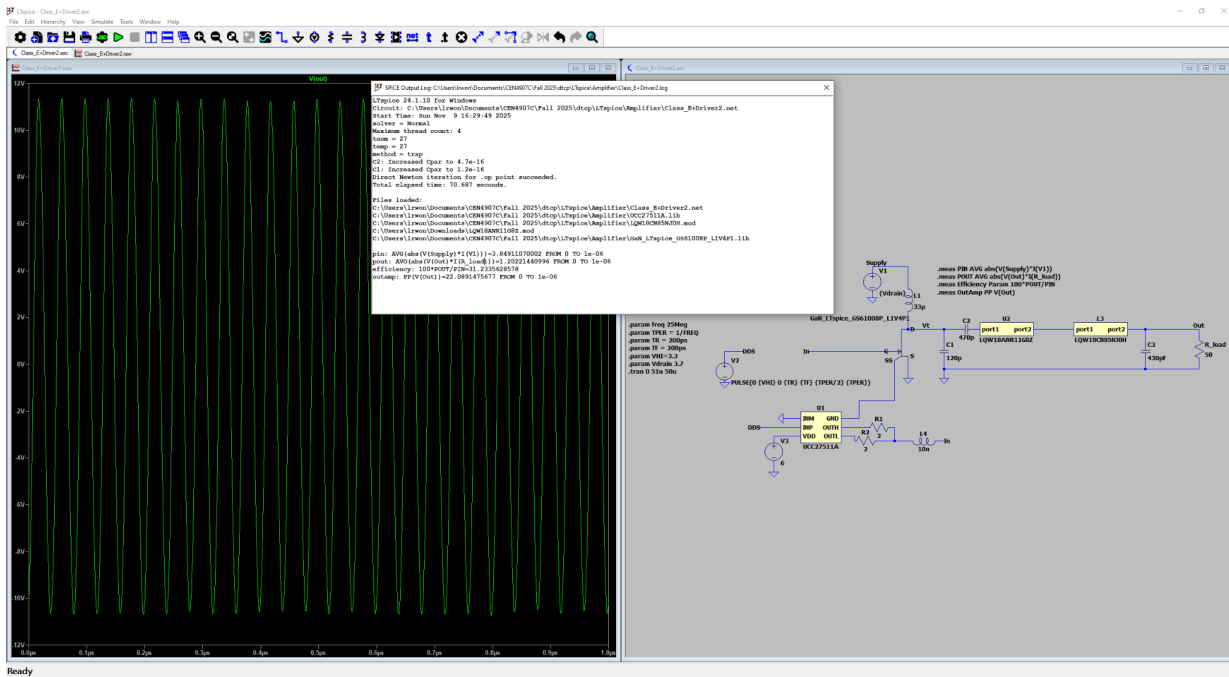


Figure 12: Once the design of the Class-E PA with real models for components to simulate, real-world performance (~31%), and switched to GaN-FET as the MOSFET was too big; found one that was bigger than the EPC2037 die and much smaller than the IRF510, the GS61008P-TR.

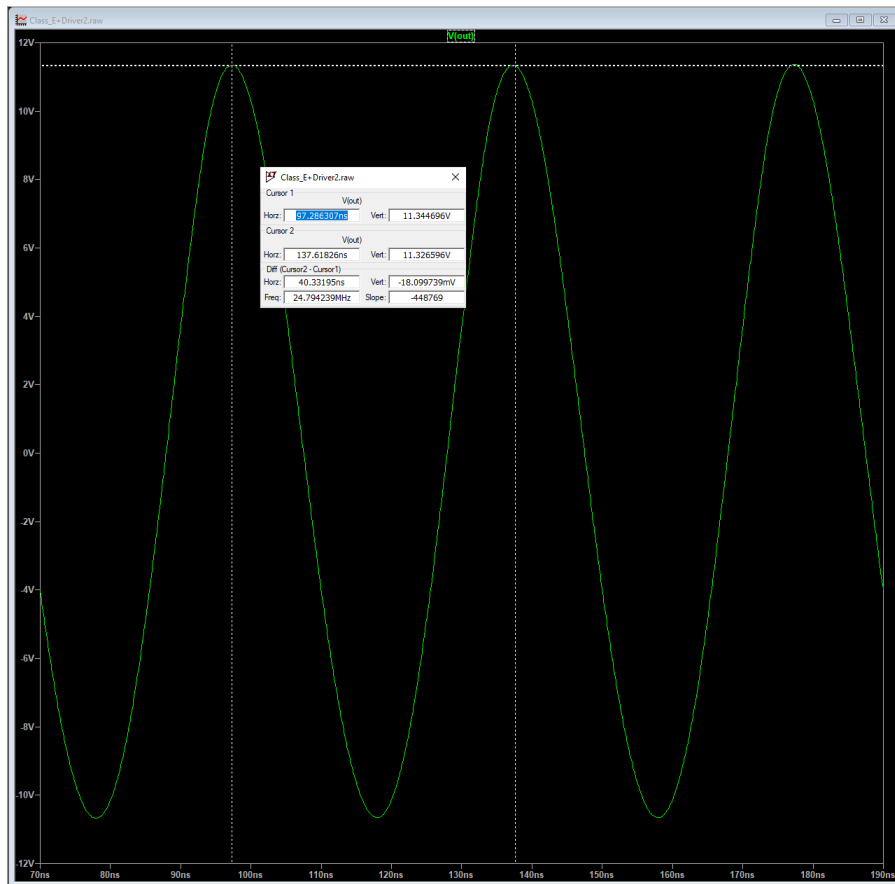


Figure 13: Confirming the device still outputs at 25MHz.

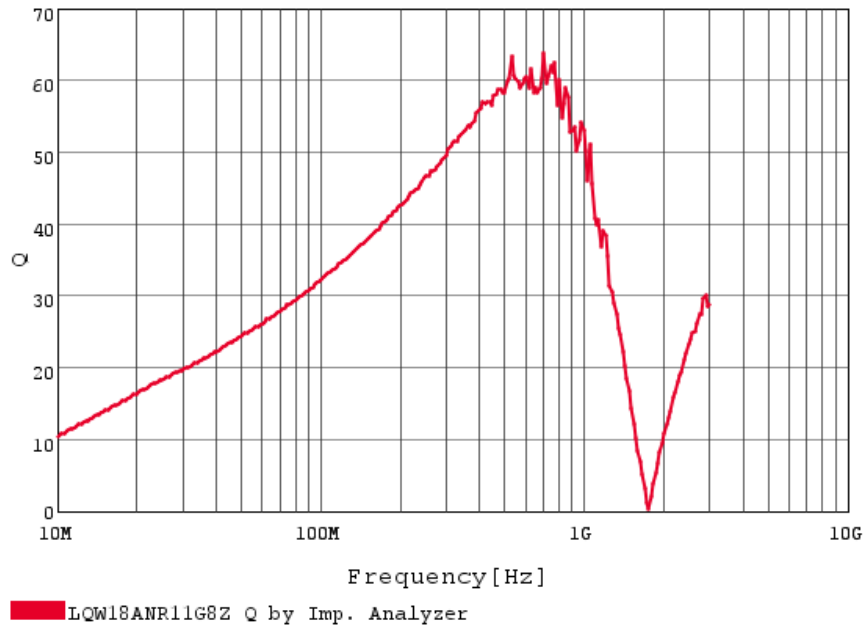


Figure 14: The spec sheet shows efficiency loss is probably due to the low Q of our inductors at our operating frequency (25MHz). We will need to test different inductors on the actual PCB once it arrives, as many components that seem to meet our needs lack a Spice model.

3. Project state summary

As of 11/15/2025, the design of the first PCB prototype has been completed, and the order was placed, with arrival of the boards expected by the end of the month. The firmware and the PA matching network are being developed and improved in parallel as we're waiting for the PCBs.

4. Project documentation

The Altium PCB project is available in this sub-directory of the main DTCP project repository:

<https://github.com/jdados/dtcp/tree/main/Altium/DTCP%20v1.2>

It can be opened by cloning the repository and then double-clicking the .prjpcb file in the File Explorer.

The firmware project is available in this sub-directory:

https://github.com/jdados/dtcp/tree/main/firmware/DTCP_Firmware

It can be opened by cloning the repository and importing the project into a local Code Composer Studio.

5. Artifact

a) External Interfaces

The board was designed for maximum debugging and measurement options, because it's meant to be used as a learning and prototyping tool to guide the next semester's final design. The user interacts with the board through numerous connectors labelled in the Altium 3D view screenshot below. There are also two indicator LEDs, where one is connected to the main 3.7 V power supply line (showing if the board is powered on), and another is controlled by a GPIO pin of the microcontroller (indicating that an EMG spike was detected).

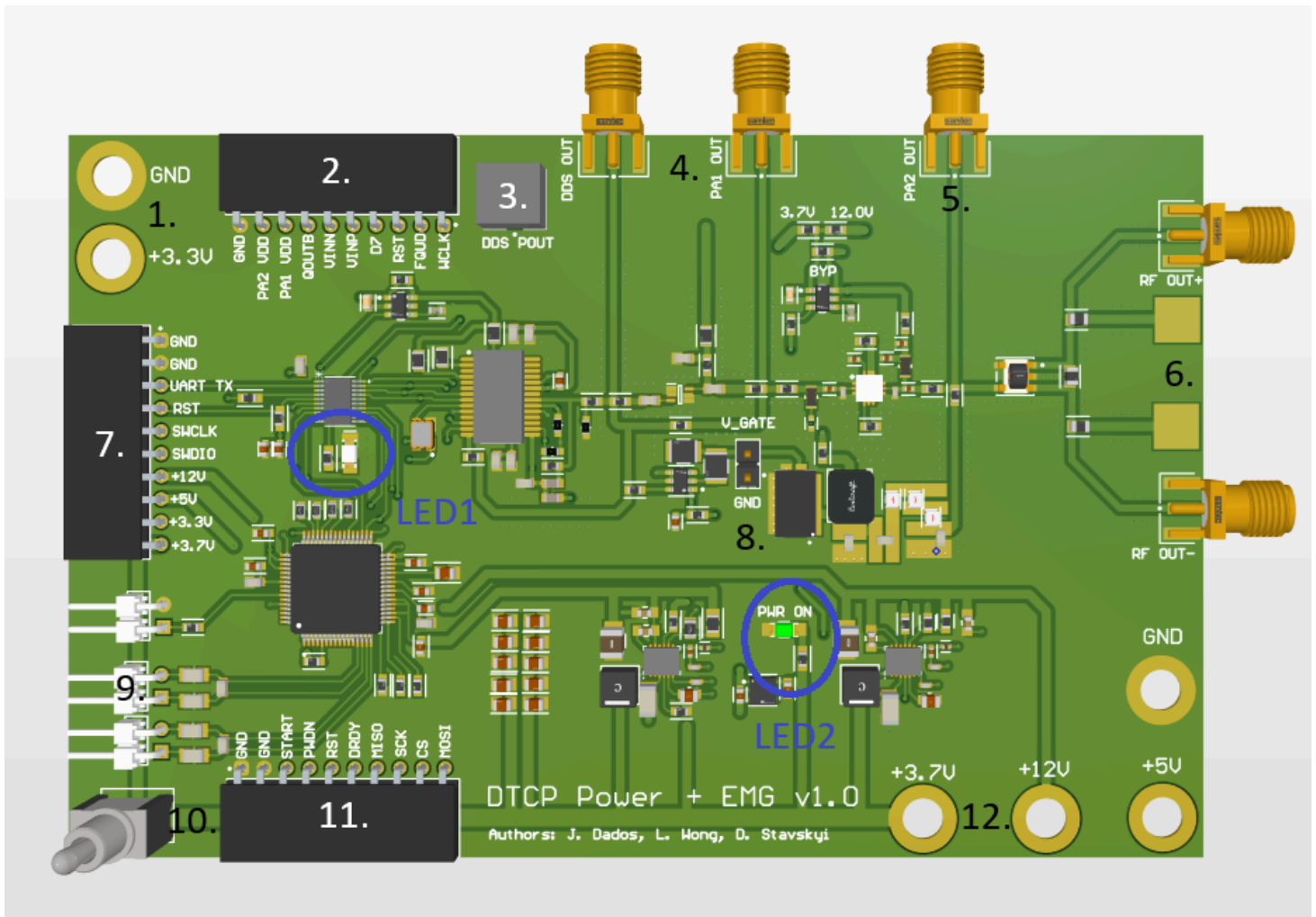


Figure 15: The Altium 3D view of the PCB.

1. Large conductive holes are used to connect a 3.3V DC power supply to the board, bypassing the LDO and directly powering the MCU, DDS, and AFE.
2. Debug header giving the user to directly measure or set the power amplifier drain DC voltages and DDS digital control signals.
3. A potentiometer sets the output power of the DDS (connected to the Rset pin of the IC).
4. SMAs to measure DDS and stage 1 PA outputs, or input signals from a signal generator to stage 2 PA when these circuits are not operational or not assembled.
5. SMA to measure the output of stage 2 PA before it's split into a differential signal by the balun. Will be used for RF performance validation.
6. The differential outputs are available through both SMAs and large pads which wires can be soldered to. This will allow for output validation on a spectrum analyzer, as well as connecting the device to external electrodes.
7. Debug header giving the user access to MCU programming signals, the UART output, and all power supply voltages (if it is desirable to access them through jumper wires over standard power supply wires).
8. For the class-E design, there is an option to directly provide the gate driving signal through a pin header in case the gate driver circuit fails or delivers low performance.

9. The EMG inputs will come from external electrodes connected to male headers on the board.
10. The main power switch is a mechanical SPST switch.
11. The third debug header provides access to all the AFE control and communication signals.
12. The second set of DC power supply connection holes for access to all device supply voltages.

b) Internal Systems

The internal hardware systems of the device are captured in this architecture diagram:

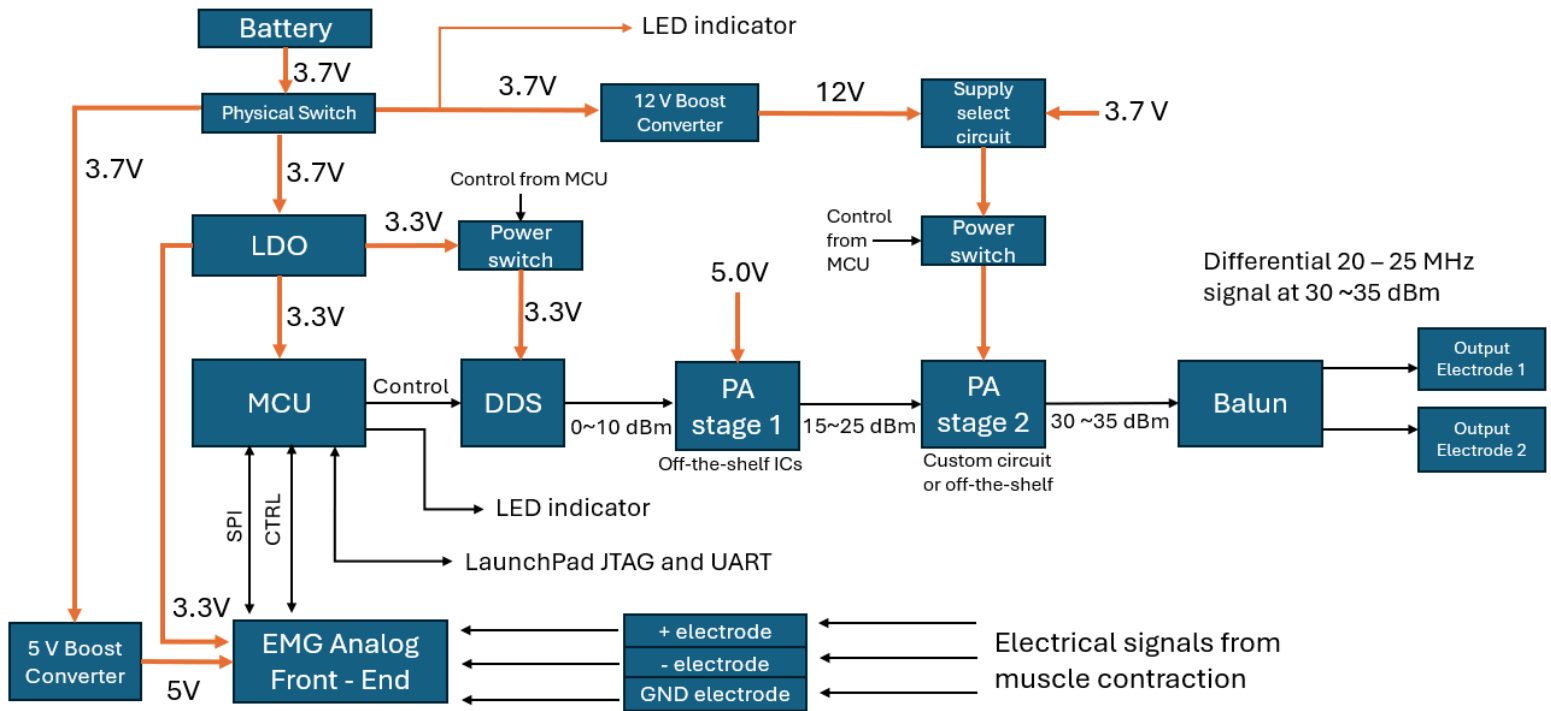
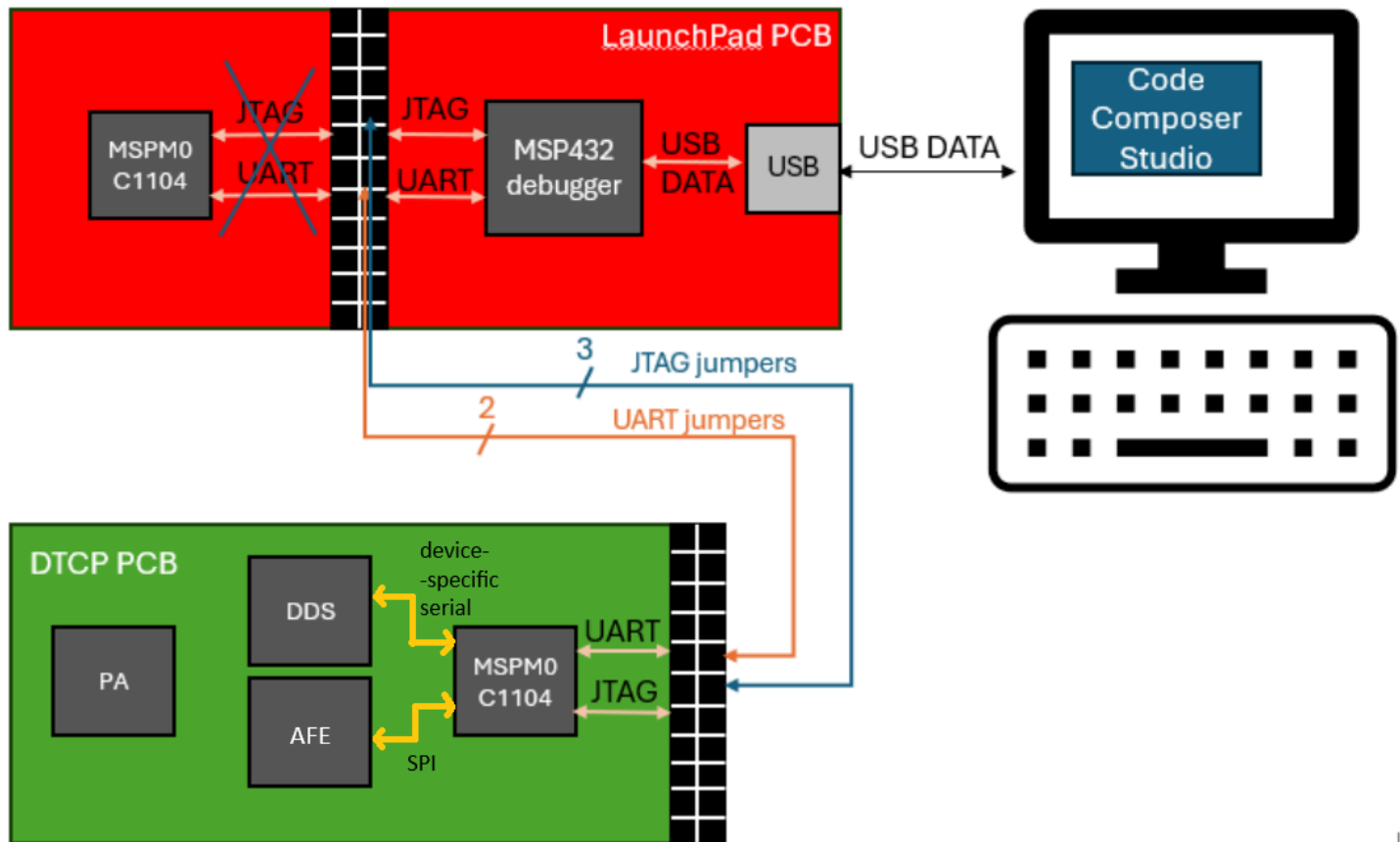


Figure 16: The system architecture diagram of the DTCP prototype.

The communication protocols used between the internal systems of the board and external devices are shown below:

Connection broken by removing the jumpers



Basic error correction will be implemented in the EMG recording firmware by validating whether the received data is valid (within the EMG signal range). Recovery will be performed by resetting the device and using the LED to indicate the device is in error state.

The PCB system-level schematic provides an in-depth view of the interactions between subsystems:

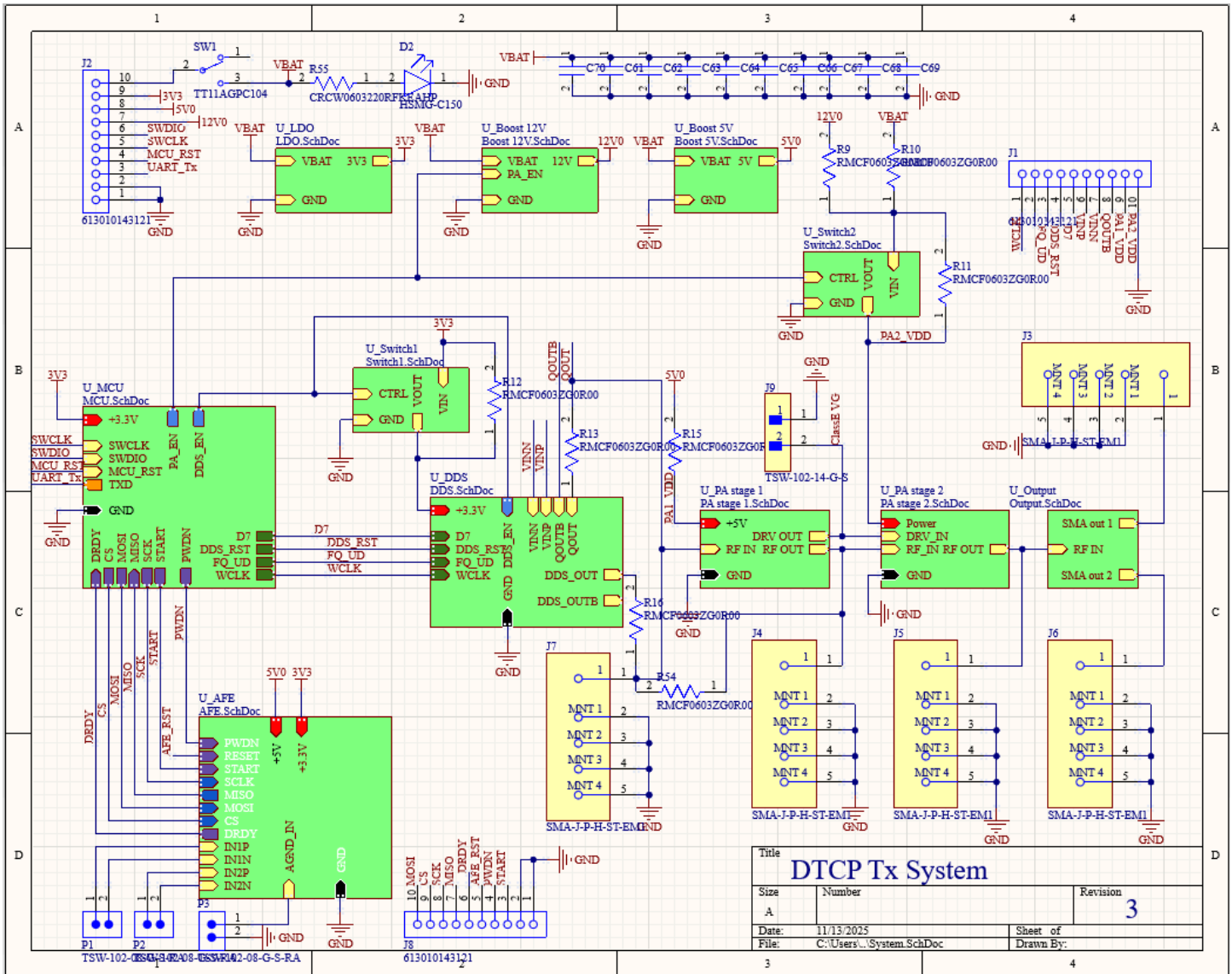


Figure 17: The system-level schematic of the prototype PCB.

As we can see, the supply select and bypassing options are all realized through a careful use of zero-ohm resistors that can create open or short circuits in the critical parts of the board.

The MCU connections had to be edited compared to pre-alpha due to the need to have an LED indicator enabling fully wireless feedback and to poll the DRDY signal on the AFE. To free two GPIOs, the UART Rx and AFE RST connections were removed, as receiving serial data from the PC is not necessary, and the AFE can be reset using an SPI command.

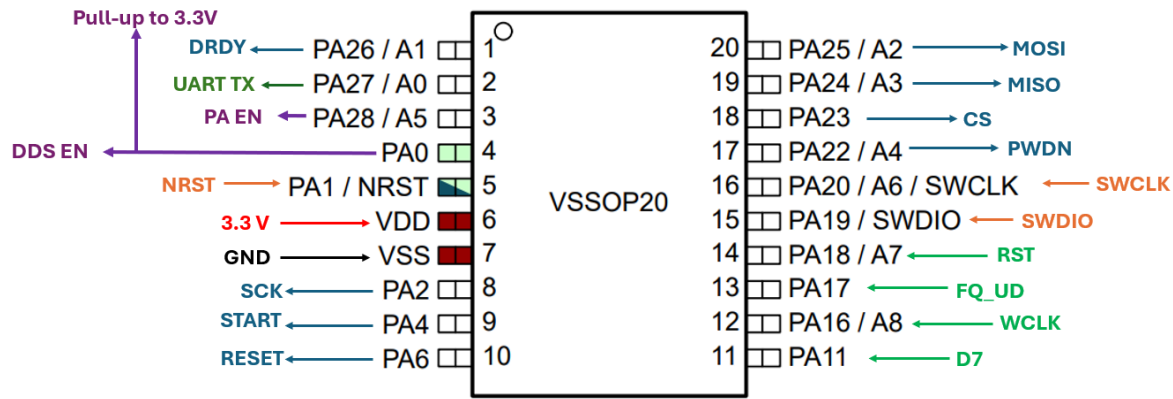


Figure 18: Connection diagram of the MCU.

The PA0 pin needs a pull-up resistor to be able to drive signals high, because it's an open-drain pin. This is the final schematic and layout of the microcontroller sub-system of the prototype:

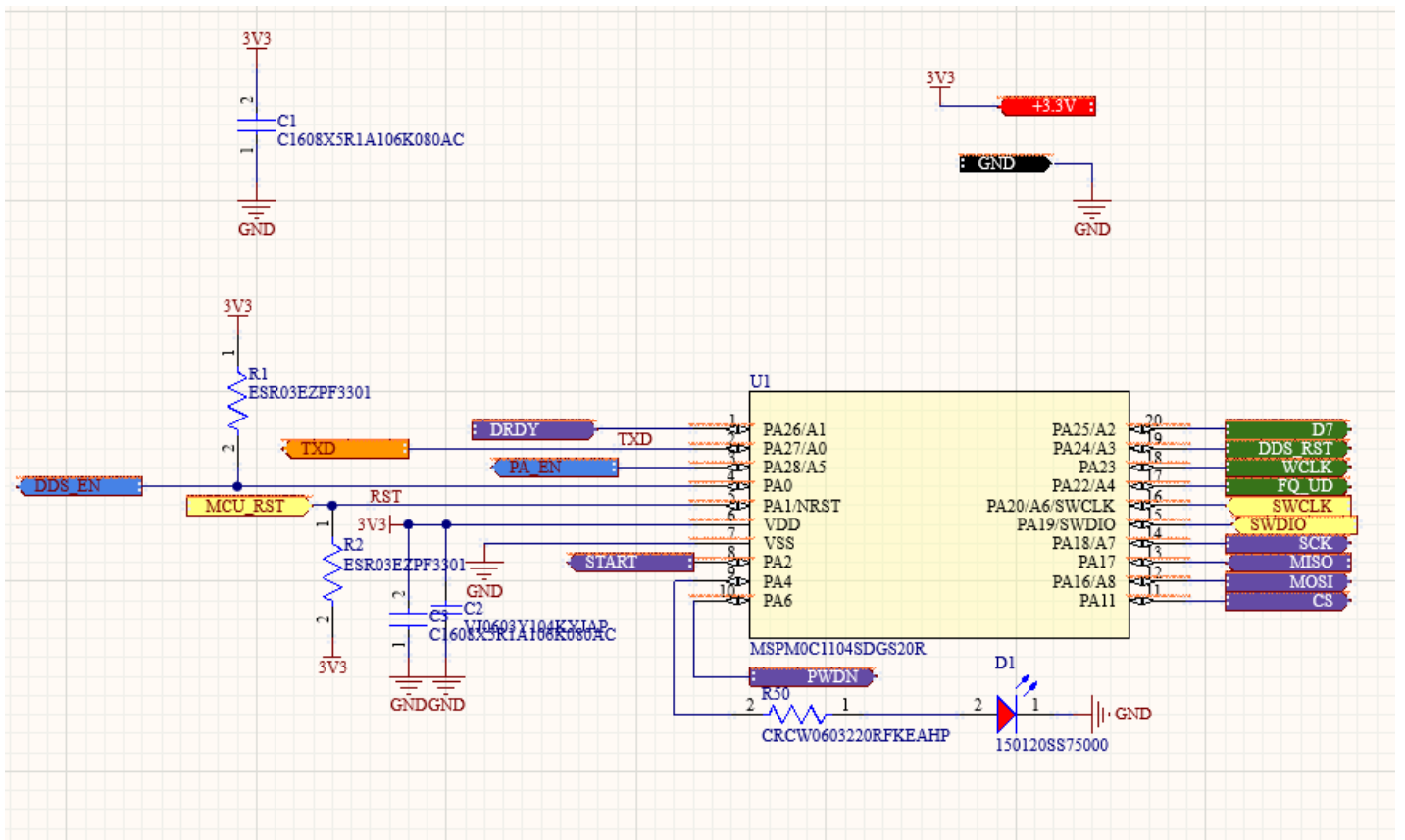


Figure 19: MCU circuit schematic.

It communicates with the DDS using a 4-pin serial AD9850-specific interface and with the AFE using SPI (4-wire) combined with 3 additional control signals (DRDY, START, PWDN). The firmware is developed using Code Composer Studio and libraries included in its Arm Cortex-M0 SDK.

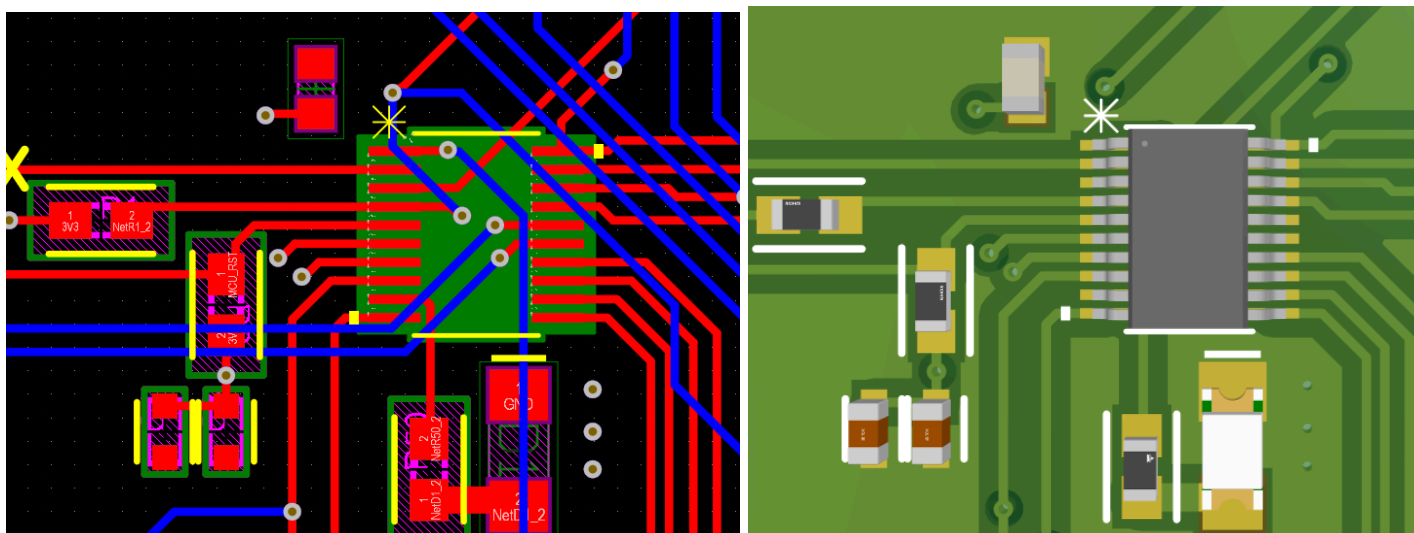


Figure 20: MCU circuit layout and 3D view.

This is the final schematic and layout of the DDS sub-system:

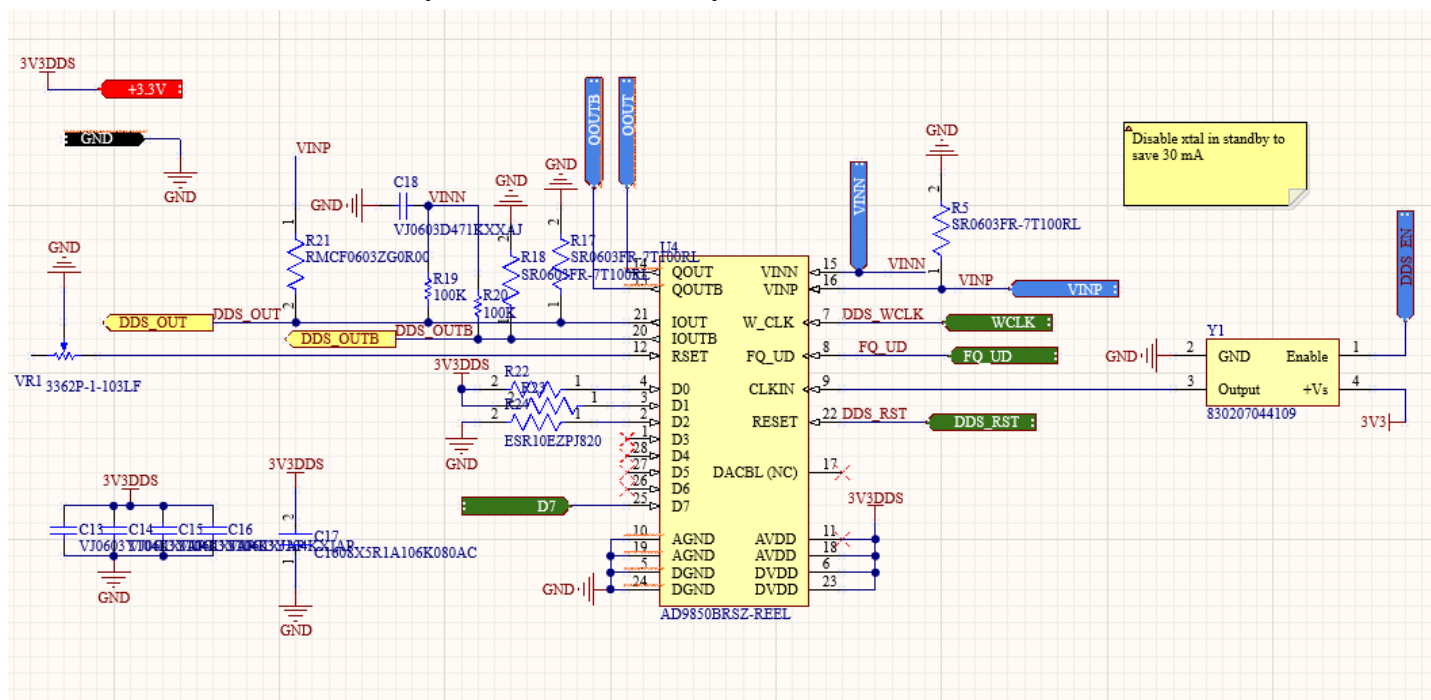


Figure 21: DDS circuit schematic.

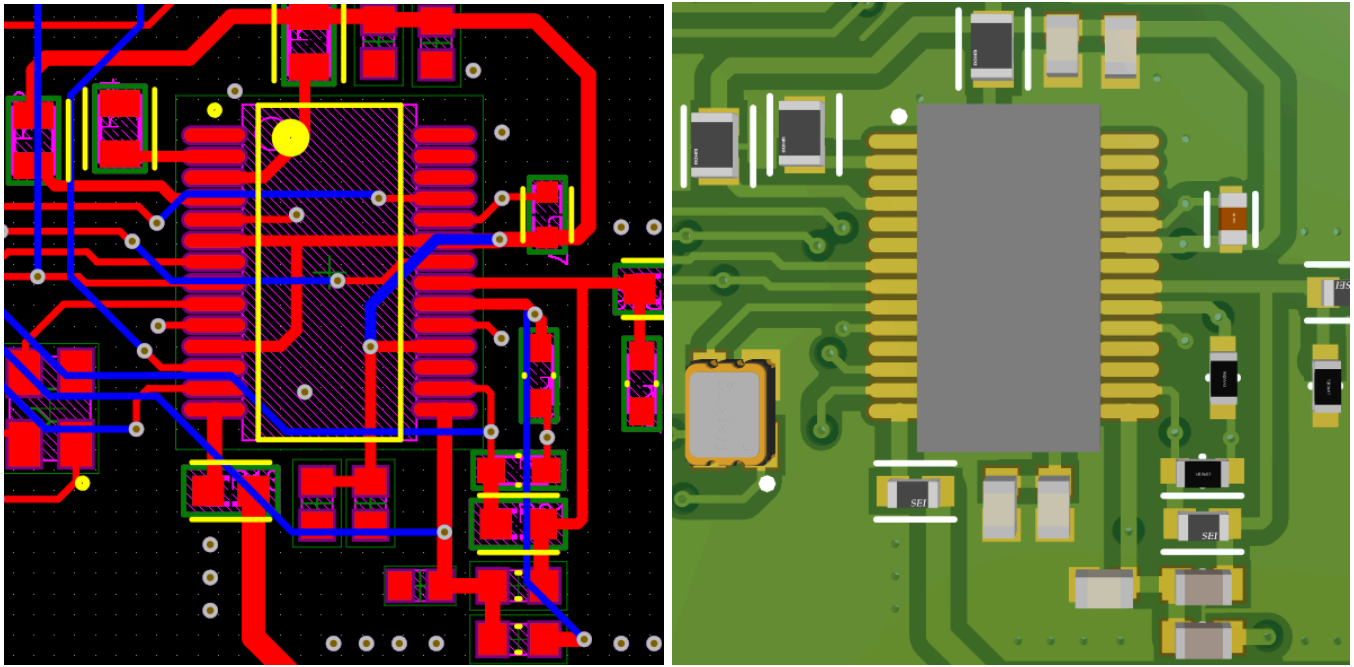


Figure 22: DDS circuit layout and 3D view.

The DDS schematic and layout support operation in two modes: sine and square wave. This is required to be able to provide inputs to both class-A and class-E power amplifiers. However, only one type of operation at once can be supported by hardware. Because of that, if only sine wave operation is required, the resistors and capacitors making up the square wave configuration circuit (in this case R5, R19, R29, and C18) will be left unassembled. Likewise, if square wave operation is required, the QOUT outputs instead of the IOUT outputs will be routed to the input of PA stage 1, and the IOUT load resistors R17 and R18 will be removed

This is the final schematic and layout of the stage 1 PA:

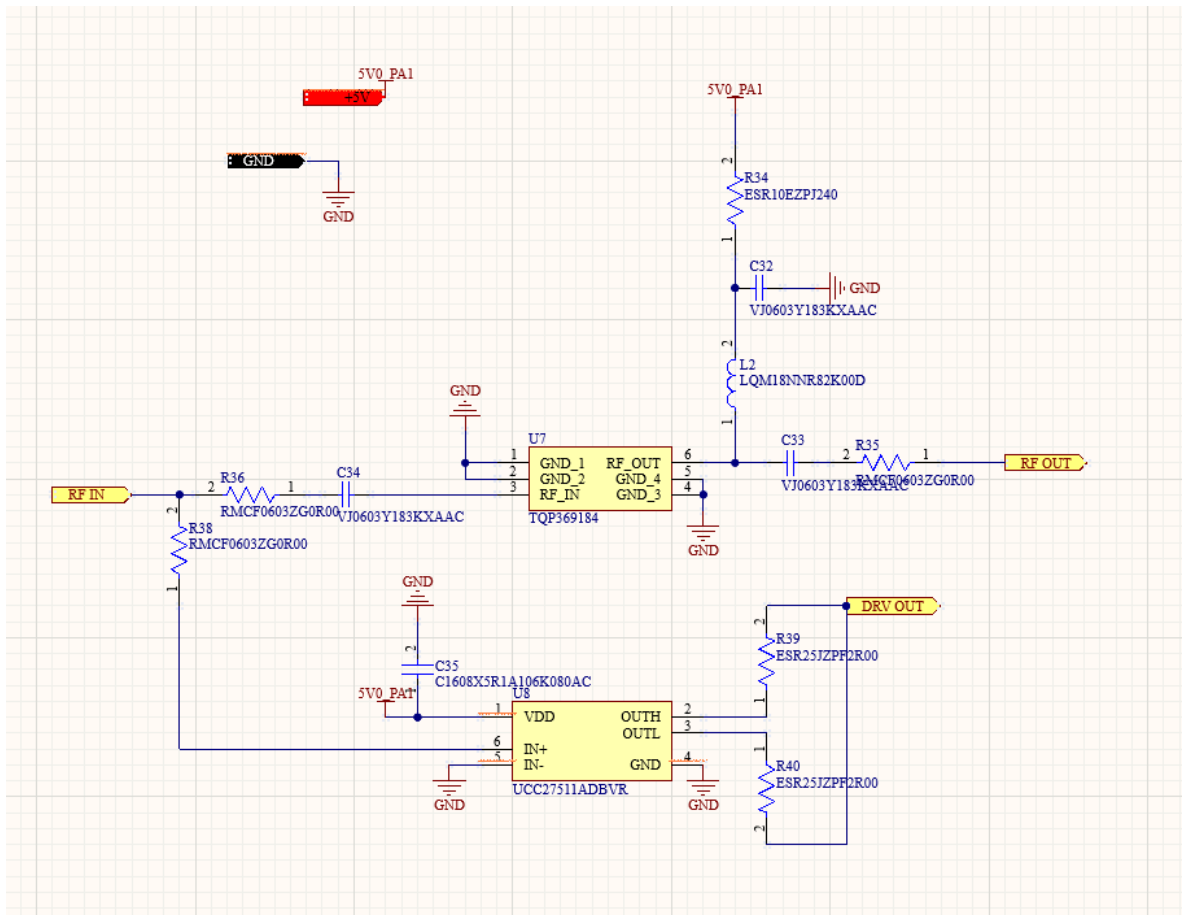


Figure 23: PA stage 1 circuit schematic.

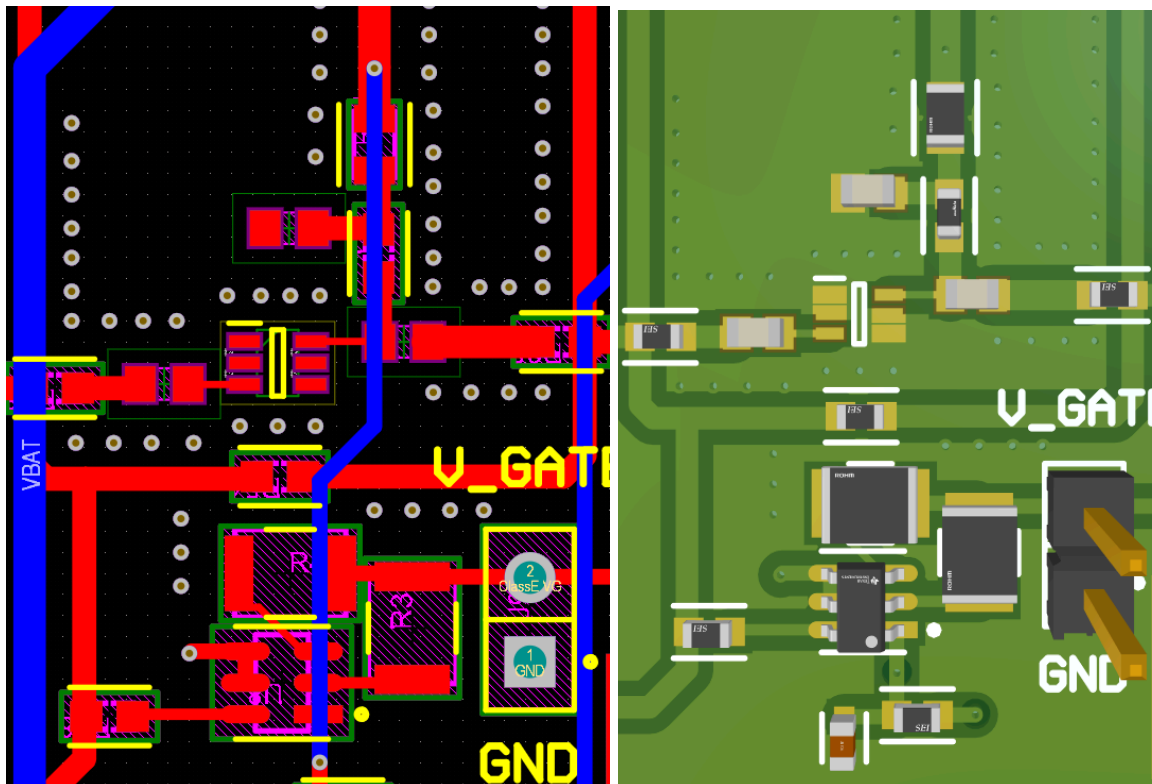


Figure 24: PA stage 1 circuit layout and 3D view.

There are two independent circuits that can be assembled, and the input is routed through two zero-ohm resistors to select which one to use. Two different ICs, TQP369184 by Qorvo with “class-A” operation (actual circuit inside the IC is unknown, but it inputs and outputs a sine wave) and UCC27511 by TI with class-E operation (square wave gate driver) will be evaluated. After the performance of class-A and class-E circuits is compared, one of them will be chosen for the second prototype. The layout also features a zero-ohm resistor, allowing the user to directly route the DDS output to the second stage if that’s desired.

This is the schematic and layout of the stage 2 PA:

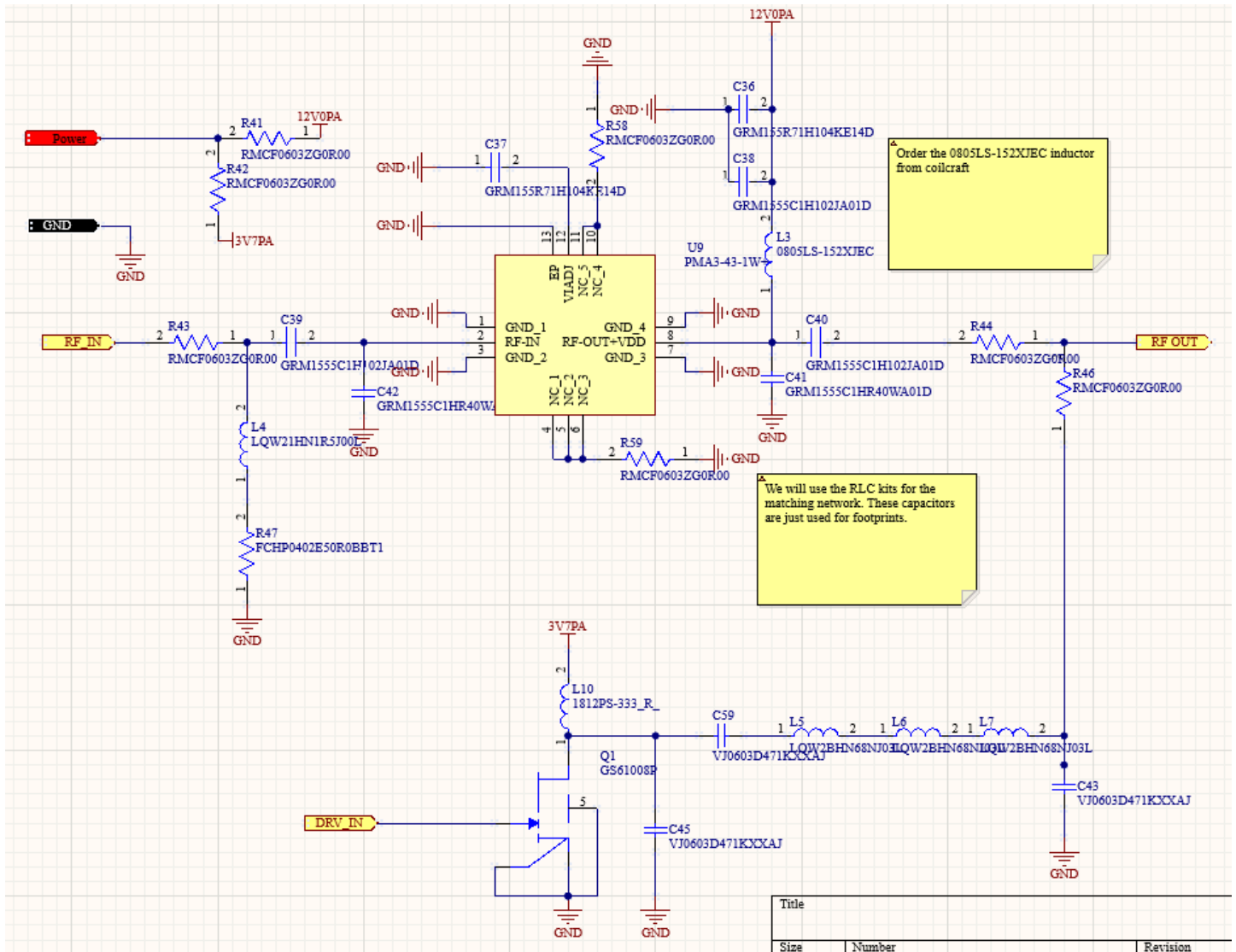


Figure 25: PA stage 2 circuit schematic.

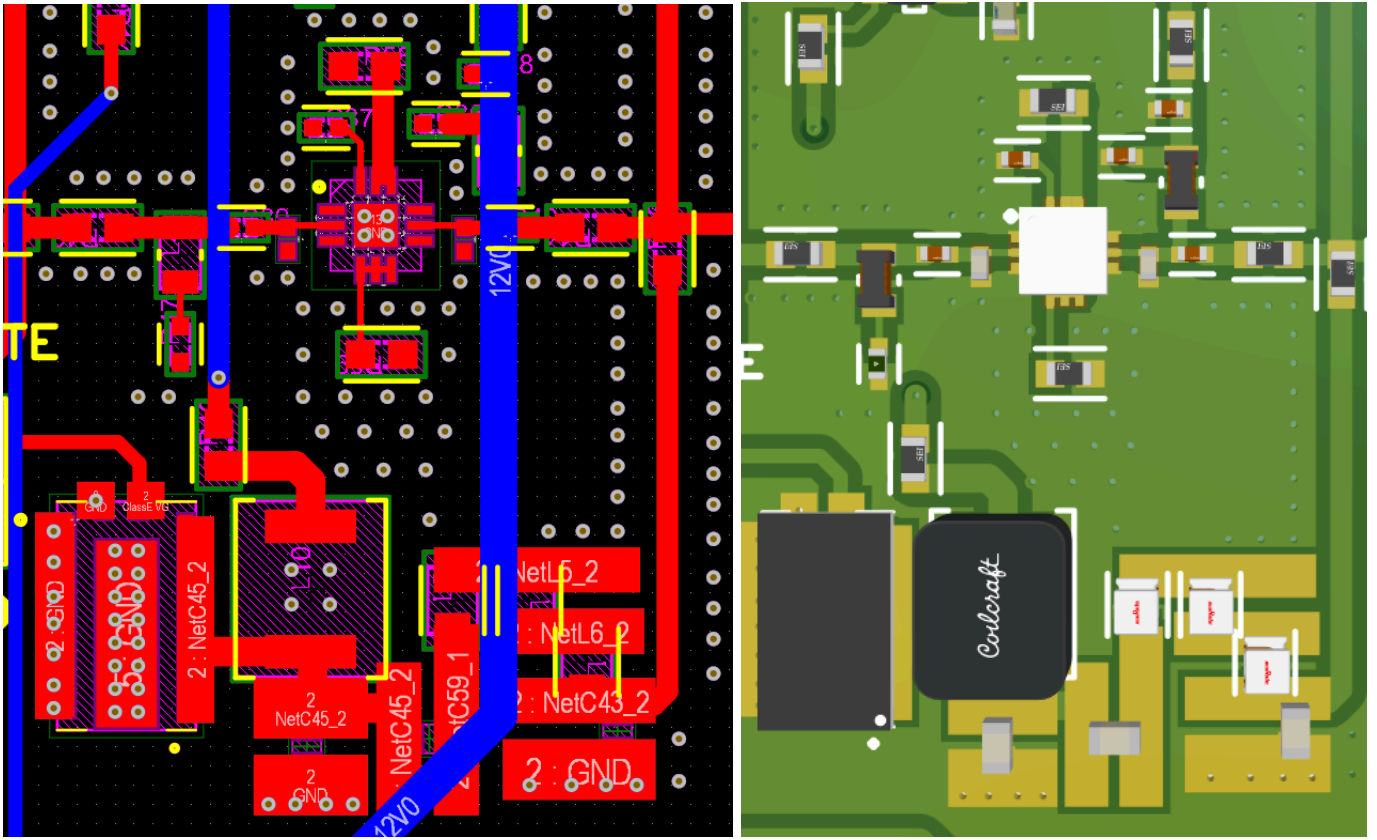


Figure 26: PA stage 2 circuit layout and 3D view.

Like in the case of the first stage, there are two independent circuits. The first one is based on a fully integrated off-the-shelf PMA3-43-12+ PA manufactured by Mini-Circuits, and the second one is our custom class-E circuit based on the GS61008P GaNFET manufactured by Infineon. This further de-risks the design by providing an alternative in the case of one circuit not working. If both circuits deliver satisfactory output power, the one with better efficiency will be chosen for the next prototype.

The pads of the class-E matching network were enlarged to accommodate multiple parallel combinations of inductors and capacitors. Because LTspice simulations could never fully capture all parasitics, the design is meant to make it easy for us to derive the optimal matching values experimentally.

The output circuit contains a 1:1 balun that splits the output signal of the second-stage PA into two differential signals, each at half of the original signal's power. There are also pads for an optional matching network that might be required between the balun and the electrodes.

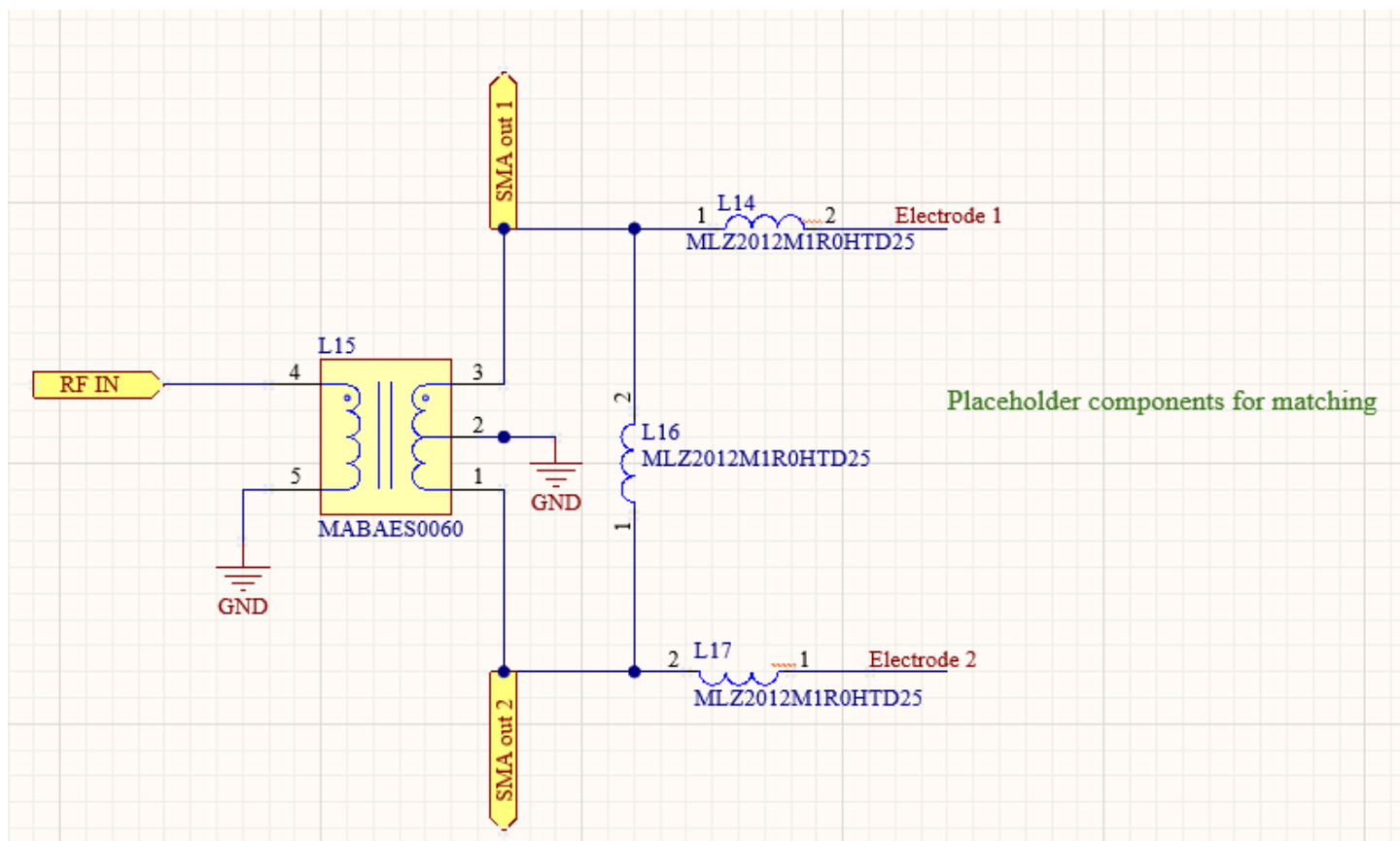


Figure 27: Output circuit layout and 3D view.

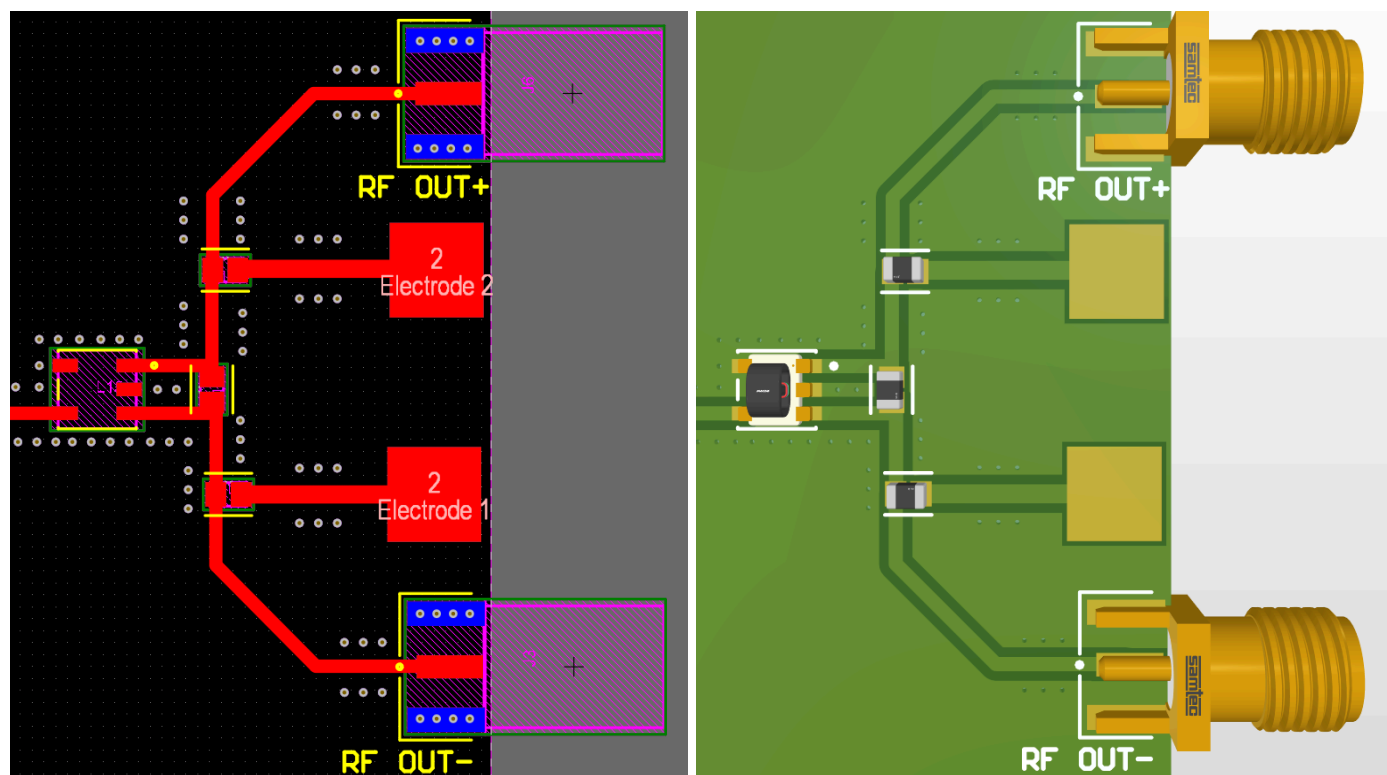


Figure 28: Output circuit layout and 3D view.

The power switches for the DDS and the second PA stage are realized using the TPS22810 IC from Texas Instruments. The same circuit is copied twice on the board with different input and output signals.

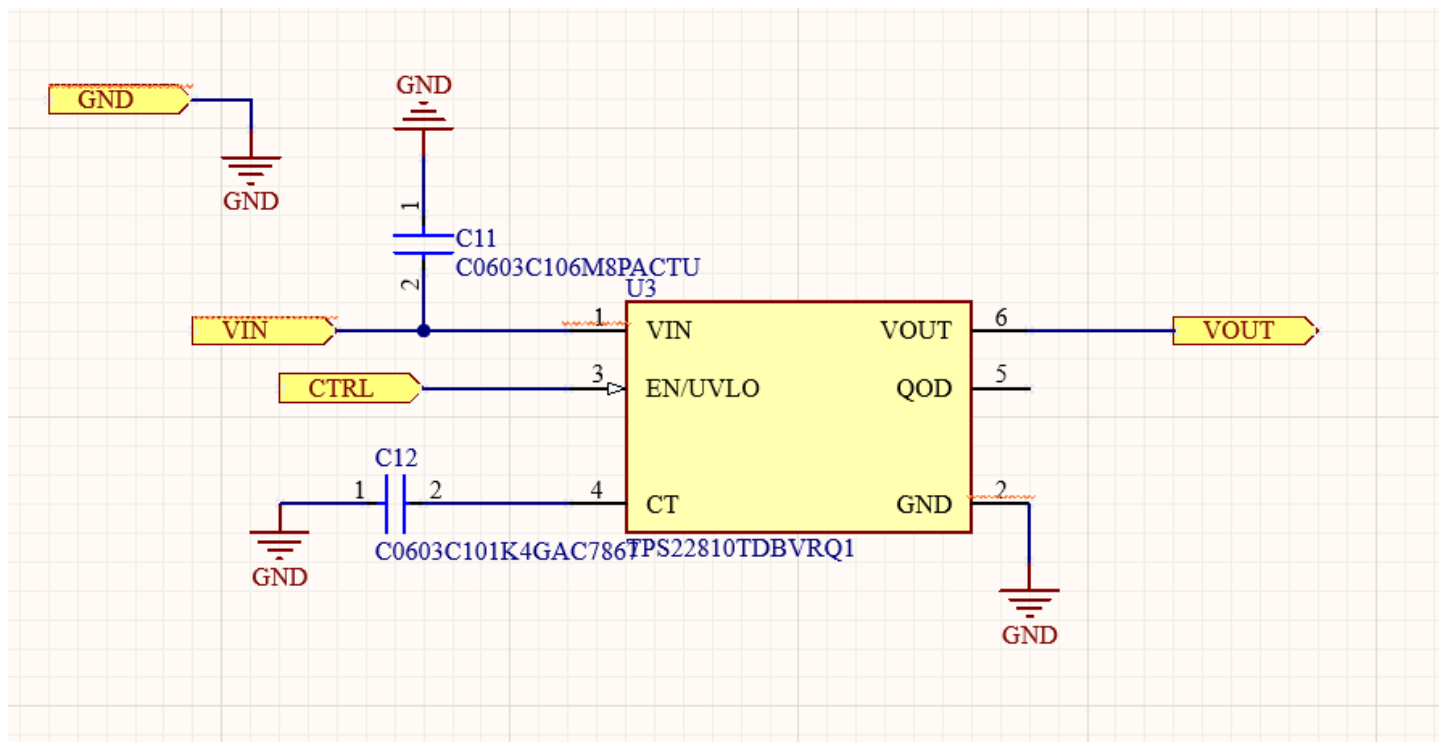


Figure 29: Switch schematic.

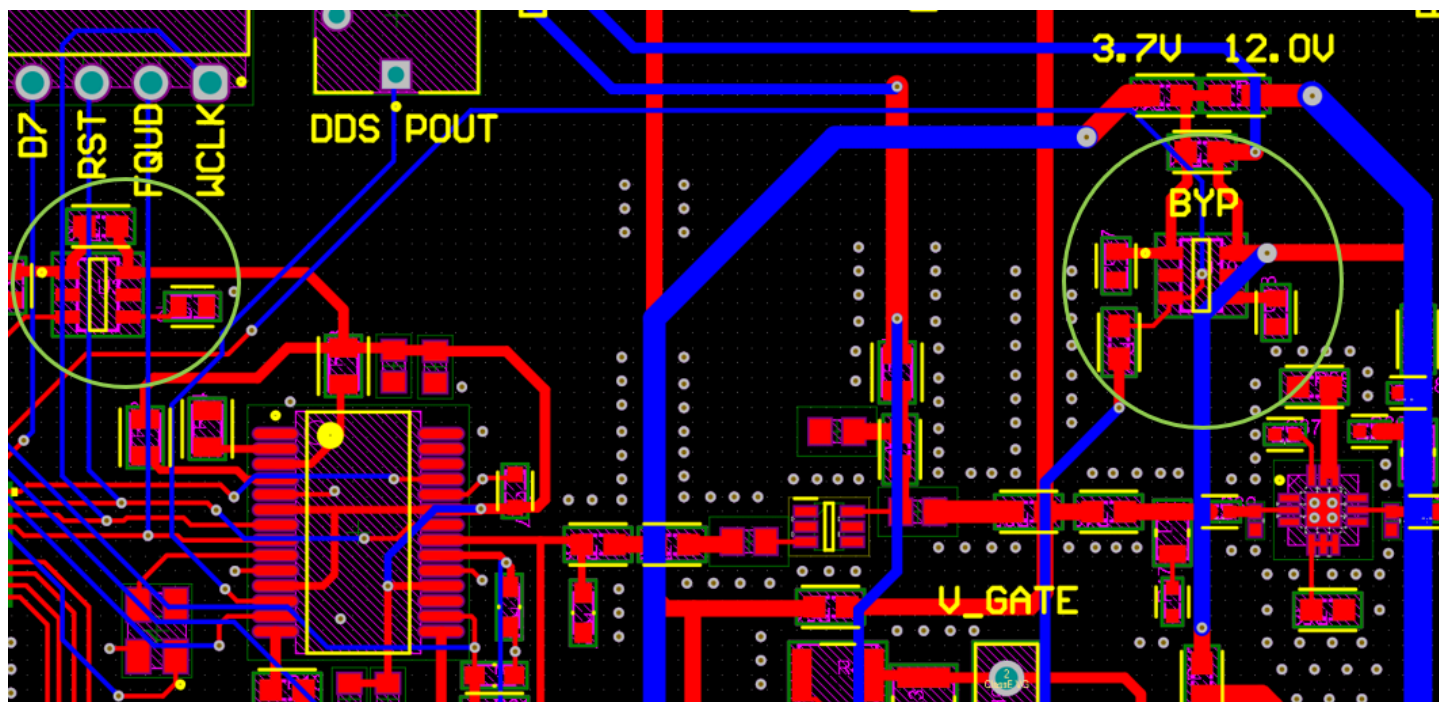


Figure 30: Switch locations marked on the layout.

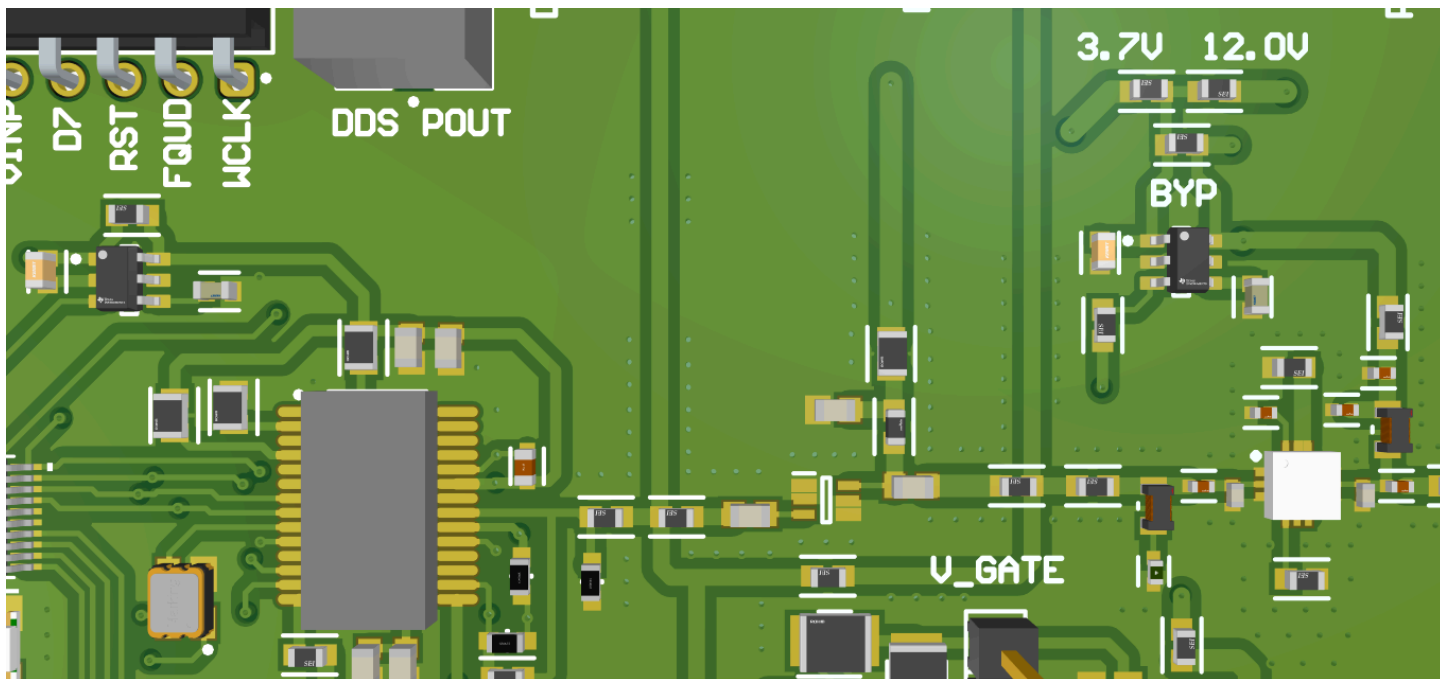


Figure 31: Switches on the 3D view.

The main power supply of the board is a 3.7 V Li-ion battery. The board power circuits use the LTC3122 boost converter by Analog Devices to generate 5.0 V and 12.0 V from that voltage. In addition, the LD39200 LDO by STM is used to drop it to 3.3 V. A bank of 10 parallel 100uF capacitors is used to provide extra current required for high-power pulsed transmission and prevent supply voltage drops.

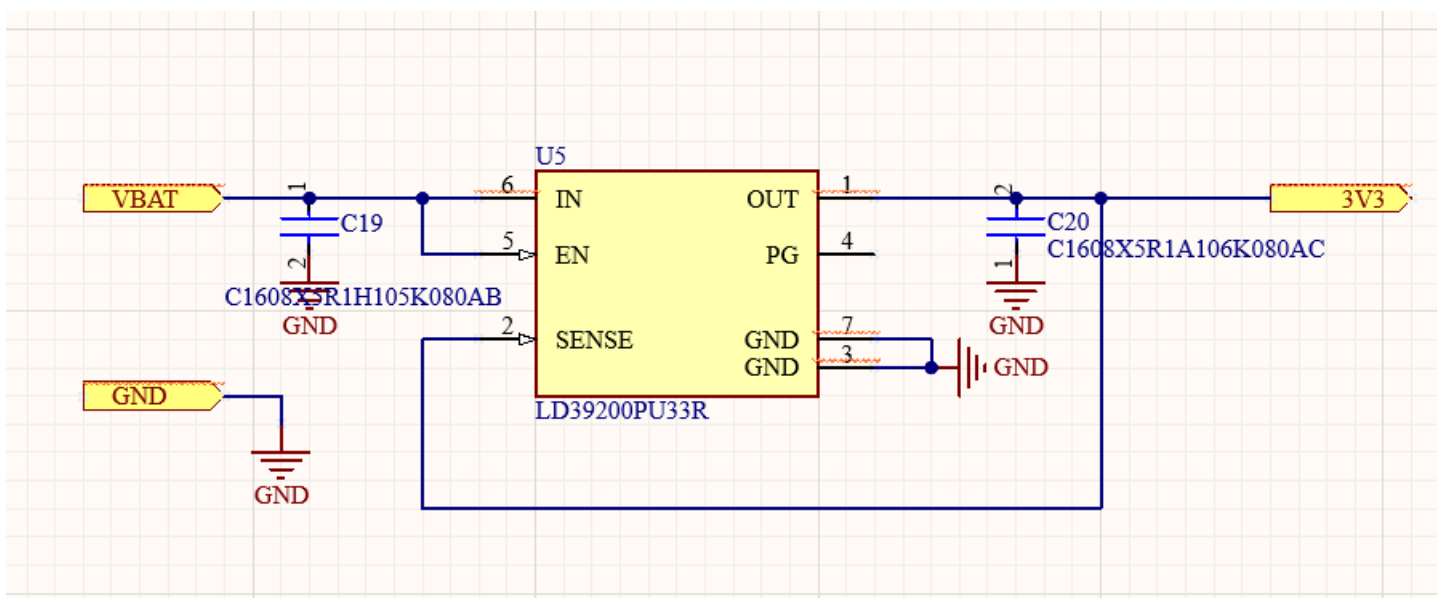


Figure 32: LDO circuit schematic.

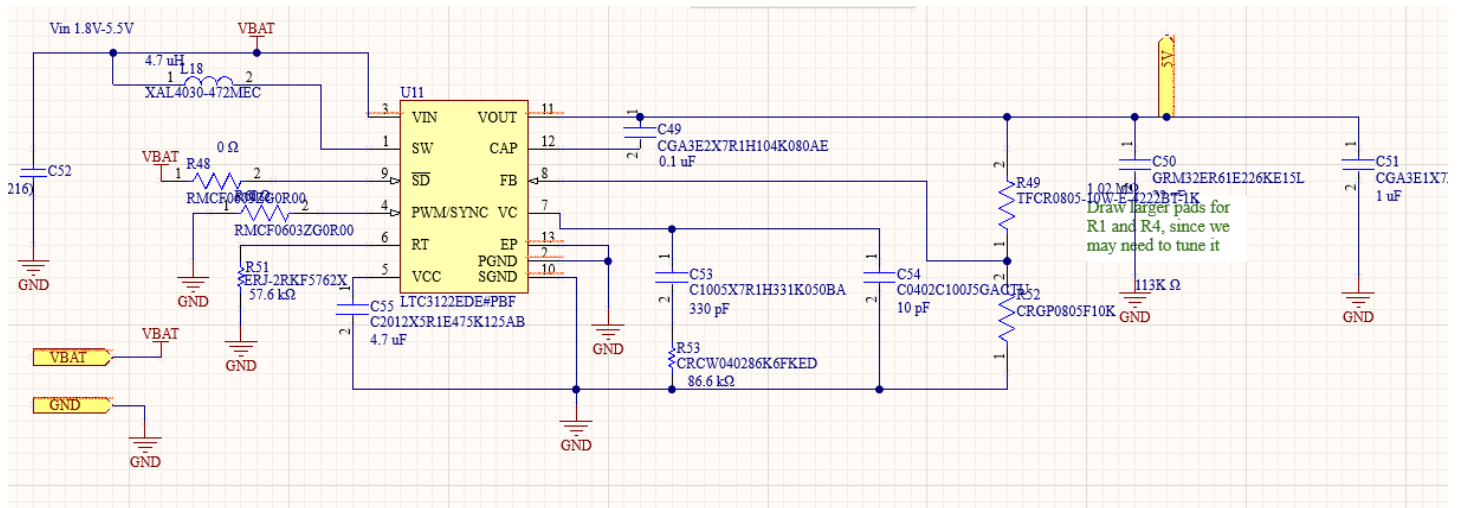


Figure 33: 5V boost converter circuit schematic.

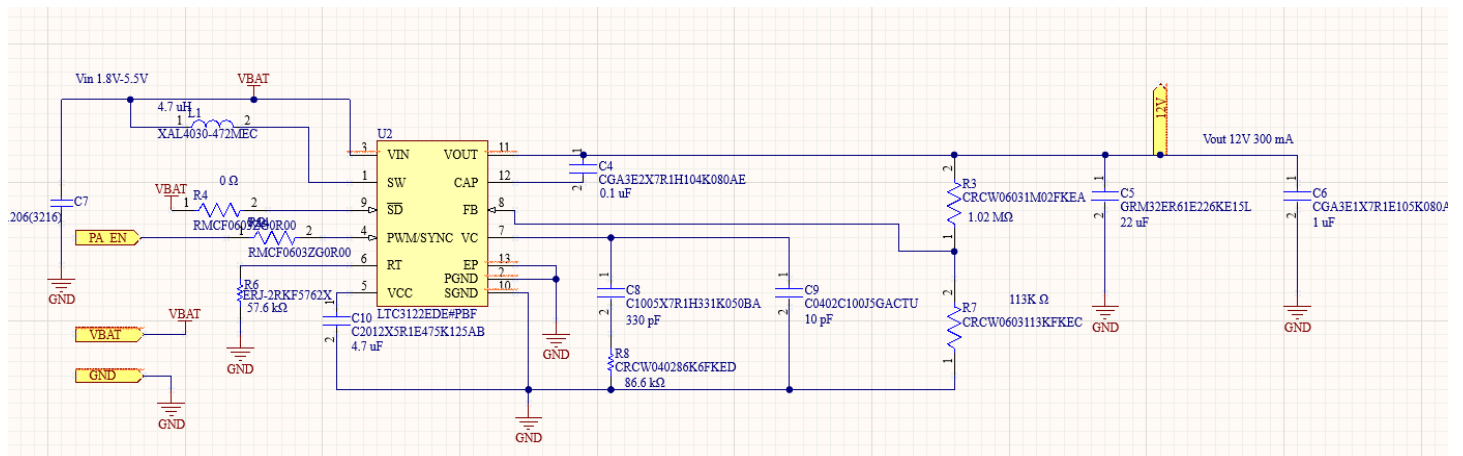


Figure 34: 12V boost converter circuit schematic.

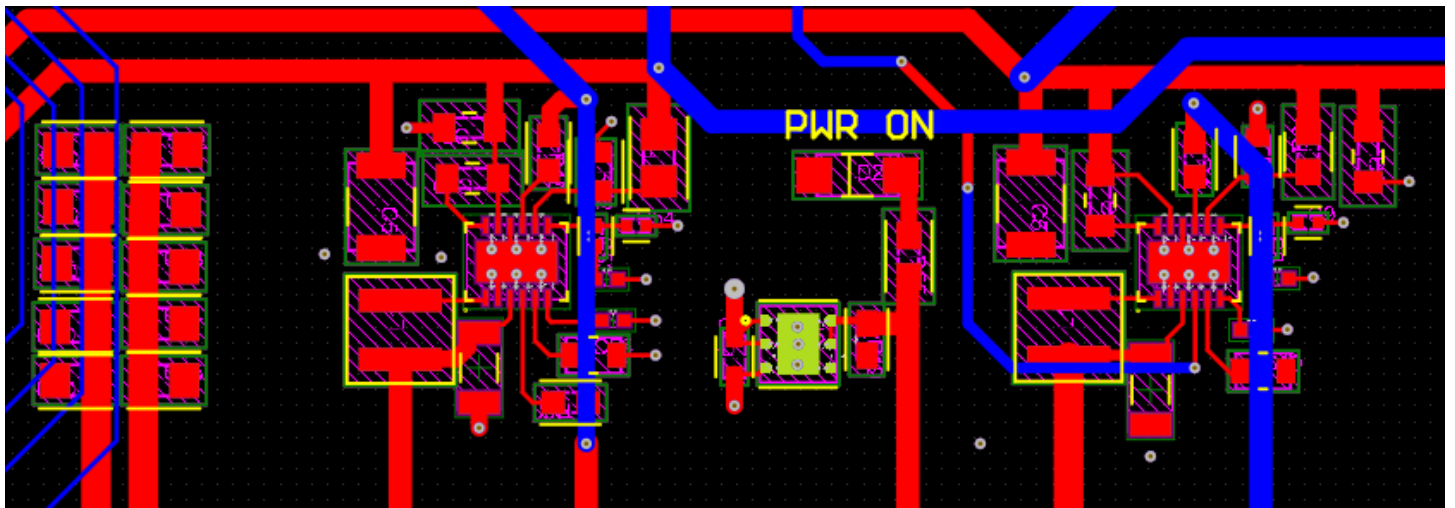


Figure 35: The layout of the board power circuits.

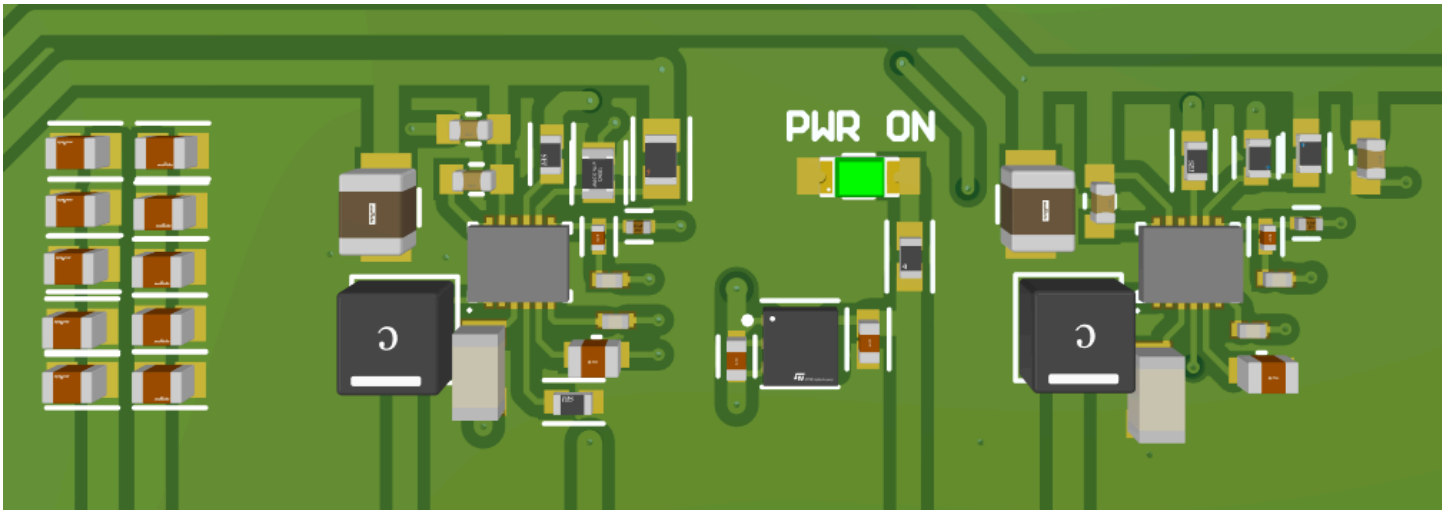


Figure 36: The 3D view of the board power circuits.

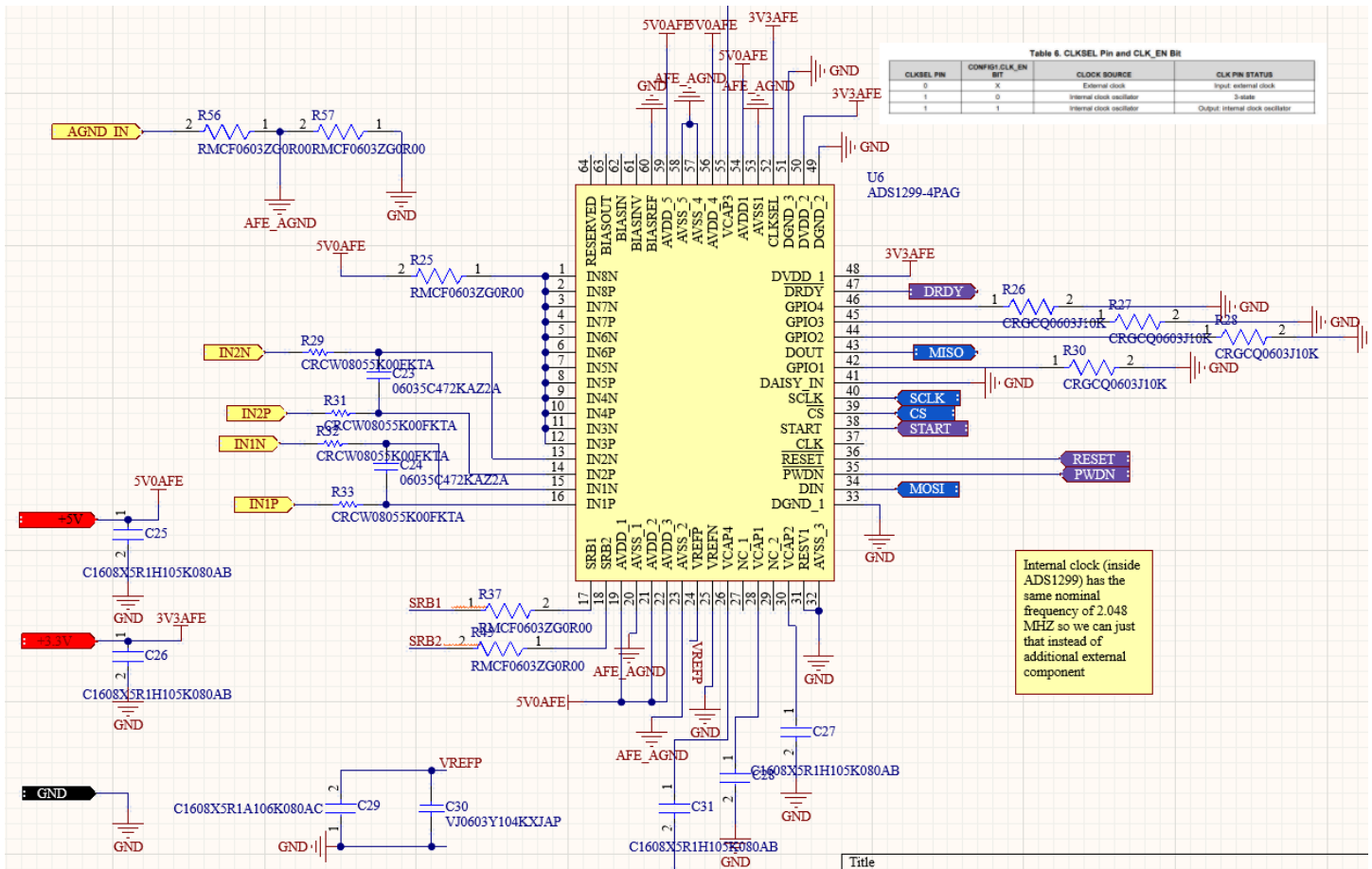


Figure 37: The layout of the board power circuits.

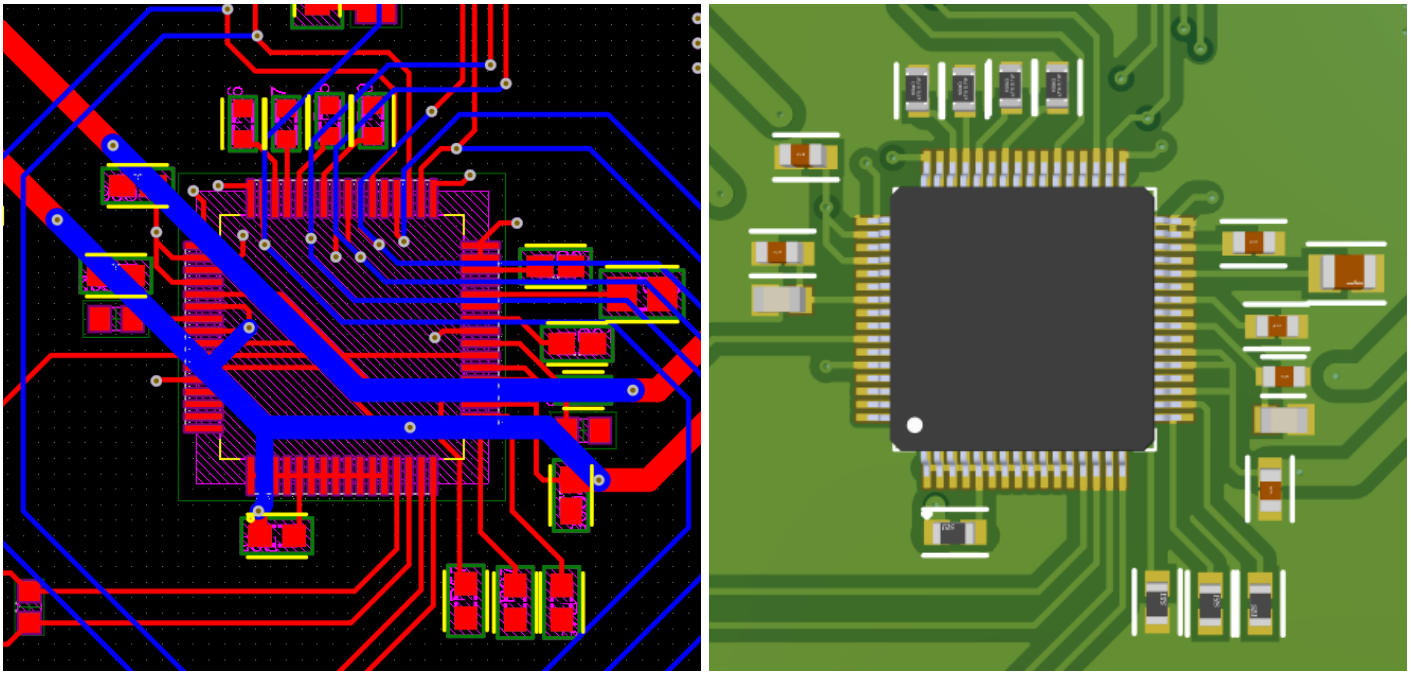


Figure 38: Layout and 3D view of the AFE circuit.

The AFE processes the signals from the electrodes, and the result is stored in shift registers, which then can be read by the MCU. It is powered by 5V for the analog part and 3.3V for the digital. The communication between the ADS1299 and the MCU and the programming of ADS1299, is done using SPI (serial communication).

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